Smart-Camp Box: An Integrated IoT and Artificial Intelligence leFramework for Safety, Communication, Psychological Support, and Resource Optimization in Disaster Relief Camps.

INDIVIDUAL THESIS

Ambalanwatta Deerasinghage Shakya Nirmani Sisikelum Deerasinghe

IT22065780

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DECLARATION

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Student name: Ambalanwatta Deerasinghage Shakya Nirmani Sisikelum Deerasinghe

Student ID: IT22065780

Signature:  Date: 10.10.2026

ABSTRACT

The immediate aftermath of natural disasters frequently disrupts critical social infrastructure, leaving millions of children in states of neurological suppression that hinder formal learning. Trauma-induced hyperactivation of the amygdala suppresses prefrontal cortex functions essential for academic retention, rendering conventional educational delivery ineffective and potentially counterproductive for displaced children [1]. Cyclone Ditwah (November 2025, Sri Lanka) illustrates this crisis: 890,000 children were affected, 1,682 schools were damaged, and teacher-to-student ratios reached 1:77, with no AI-driven educational systems deployed in the emergency response [2]. Existing educational technologies assume learners are psychologically prepared to engage with content; no system currently assesses combined psychological and attentional readiness before instruction or monitors individual children for persistent welfare distress [3],[4].

This study introduces the Psycho-Attentional Gated Educational System (PAGES), an offline-first IoT web application designed to assess readiness among Grade 3-4 children (ages 8-9). PAGES uses a three-source behavioral fusion model that integrates a gamified inhibitory control task, passive interaction signals, and a rolling historical baseline. A binary gate decision determines access to standardized curriculum content per session. A two-phase anomaly detection pipeline applies Z-score statistical thresholding followed by per-child Isolation Forest modeling to monitor behavioral patterns and generate plain-language welfare alerts for admin.

The system uses a MERN stack extended with Python and scikit-learn, SQLite for offline edge storage, and is deployed on Raspberry Pi 5 hardware without internet dependency. A pilot study with 29 students from Cyclone Ditwah-affected schools under the Homagama Zonal Education Division is expected to demonstrate significant assessment sensitivity ($p < 0.05$), gate accuracy above 80%, curriculum alignment exceeding 90%, welfare alert precision of 0.75, and recall of 0.70, and complete offline functionality with full cloud synchronization fidelity. Another pilot study carried with not affected children from Hiramadagama Maythree Maha Vidyalaya. Future work will evaluate PAGES in active emergency deployments, extend anomaly detection with clinical validation, and integrate complementary Smart-Camp Box components.

Keywords: *post-disaster education, psycho-attentional gating, anomaly detection, Isolation Forest, offline IoT, child welfare monitoring*

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LIST OF ABBREVIATIONS

posttraumatic stress disorder - PTSD

Psycho-Attentional Gated Educational System - PAGES

National Institute of Education - NIE

United Nations Educational, Scientific and Cultural Organization - UNESCO

United Nations Children's Fund - UNICEF

National Disaster Management Centre - NDMC

Zonal Education Office - ZEO

Zone of Proximal Development - ZPD

Continuous Performance Test - CPT

Organisation for Economic Co-operation and Development - OECD

Information Technology Services - ITS

United Nations - UN

Artificial Intelligence - AI

Internet of Things - IoT

Design Science Research - DSR

1. INTRODUCTION

1.1 Overall project context

Sri Lanka's escalating exposure to climate-driven disasters 304 climate-related events between 2015 and 2022, compared to just 35 between 2008 and 2014 has repeatedly exposed a structural weakness in national relief infrastructure: camps established to protect displaced populations are under-monitored, poorly connected, and unequipped to support the psychological and educational needs of the children sheltering within them. Cyclonic Storm Ditwah, which made landfall on Sri Lanka's eastern coast on 28 November 2025, brought this weakness into sharp focus at national scale, displacing 1.4 million people, damaging 1,682 schools, and affecting more than 890,000 children [1]. At the peak of the emergency, floodwater inundation forced 894 of 990 active relief camps to close, while communication network failures collapsed manual identity-verification and resource-distribution processes precisely when they were needed most. Concurrently, a documented teacher-to-student ratio of 1:77 and 28,900 trauma-affected teaching staff left displaced children without any psychologically informed educational support, despite neuroscientific evidence that unresolved trauma suppresses the prefrontal-cortex functions required for learning.

This project addresses these interconnected gaps through the Smart Camp Box, a unified, offline-first IoT and AI platform developed collaboratively across three integrated components: Smart Resource Management and Semantic-Aware Message Prioritization (SAMP), developed by Kashmira S S; Smart Environmental Risk Monitoring and Location-Aware Flood Prediction, developed by Hansana G P; and the Psycho-Attentional Gated Educational System (PAGES), developed by the author of this thesis. Each component addresses a structurally distinct subproblem: physical camp safety, resilient identity and resource communication, and trauma-informed education, while sharing a common deployment architecture, local data layer, and cloud-synchronized analytics extension, so that the three function as one coherent system rather than three unrelated tools. The overall project context, shared architecture, and cross-component integration are documented in full in the group's common report; this thesis presents the detailed research, design, development, and evaluation of the PAGES component specifically.

1.2 Individual Component Context

This thesis addresses the third of the project's three subproblems: the delivery of educational content to displaced children without regard for their psychological or attentional readiness to receive it. Existing humanitarian education technologies, including offline platforms such as Kolibri, KA Lite, and RACHEL, as well as static resources such as UNICEF learning kits and Save the Children digital packs, operate on the implicit assumption that a child present in a learning environment is cognitively available to engage with academic content. This assumption is neurologically unfounded for acutely traumatized children: sustained activation of the amygdala's threat-detection circuitry during post-traumatic stress suppresses prefrontal cortex activity, the neural region responsible for attention, working memory, and the executive functions that learning requires [2]. A child in acute post-disaster distress may therefore be

physically present in an educational setting yet remain neurologically unavailable to learn, regardless of the quality or appropriateness of the content offered to them [4].

PAGES addresses this gap through a psycho-attentional gating architecture, in which a child's combined psychological and attentional readiness, measured passively through behavioral interaction signals rather than self-report, determines whether they are permitted to engage with standardized cognitive educational content at all [5]. This is a deliberate departure from adaptive learning systems such as DreamBox and Knewton, which adjust content difficulty based on observed academic performance; PAGES instead adapts the gate, using a composite readiness score drawn from a gamified Go/No-Go attentional task (50%), passive behavioral signals (30%), and a rolling historical baseline (20%) to decide whether cognitive content is released at all, while the content itself remains standardized across all children who pass the gate. This architecture is grounded in Van der Kolk's trauma neuroscience, Blair and Razza's evidence that self-regulation is a biological prerequisite for learning rather than a facilitating factor, and the sequencing principles of Maslow's hierarchy of needs, Vygotsky's Zone of Proximal Development, and Bloom's Mastery Learning model.

Within the shared Smart Camp Box architecture, PAGES operates as one of three edge modules running on the device's local WiFi network, persisting session data to its own local SQLite database and synchronizing anomaly-flagged sessions to the cloud analytics layer developed as this component's own extension: a two-phase machine learning anomaly-detection system (Z-score from session three, Isolation Forest from session eight) that identifies children exhibiting persistent psycho-attentional distress across multiple sessions and generates welfare alerts for camp administrators. The remainder of this thesis presents the detailed literature review, methodology, implementation, and evaluation of PAGES, including its pilot deployment with 29 Grade 3-4 children across 580 sessions at Mullegama Maha Vidyalaya and Jalthara Maha Vidyalaya.

1.3 Problem Definition and Contextual Framing

The intersection of large-scale natural disasters, forced displacement, and psychological trauma among children represents one of the most complex challenges that remains insufficiently addressed in contemporary humanitarian technology design. When Cyclone Ditwah made landfall on Sri Lanka’s eastern coast on November 28, 2025, it produced not merely a physical emergency but a cascading educational and psychological crisis affecting an estimated 890,000 children across nine provinces [1]. In a period of just 72 hours, 1,682 primary and secondary schools were damaged or rendered structurally unsafe, displacing thousands of children into temporary relief camps. In these settings, the foundational conditions for learning, psychological safety, cognitive availability, and access to supportive learning structures were not only absent. Still, they were a predictable outcome of mass migration due to natural disasters [2].

The challenge of post-disaster education is not purely logistical. It is deeply psychological and attentional in nature. Children affected by disaster exhibit clinically significant levels of post-traumatic stress, anxiety, grief, and emotional dysregulation [3]. These psychological

states directly impair cognitive functioning, attentional capacity, and learning readiness, meaning that even when educational resources are physically restored, a traumatized child may be present in a learning environment yet remain neurologically and psychologically unavailable to learn [4]. Neuroscientific evidence confirms that chronic stress responses in young children suppress prefrontal cortex activity, the brain region responsible for attention, working memory, and learning, making cognitive engagement physiologically impossible under conditions of unresolved psychological distress [5].

Many children exposed to traumatic events develop distress symptoms, which can represent normal reactions to abnormal circumstances [6]. These reactions may include fears of recurrence, anxiety, sadness, grief, sleep disturbance, difficulty concentrating, anger, behavioral outbursts, or developmental regression [7],[8]. Increases in risk-taking behaviors, such as alcohol and substance use, have also been described among adolescents after disasters [9]. Some children will experience symptoms that meet criteria for psychiatric disorders, such as acute stress disorder and posttraumatic stress disorder (PTSD).

Evidence is built on the development of posttraumatic stress symptom profiles and trajectories in children exposed to disasters. A study of eight studies on posttraumatic stress symptom profiles in children exposed to natural disasters found that between 4% and 23% had a chronic posttraumatic stress symptom profile, including elevated levels of PTSD over time. An additional 7% to 27% had a recovery profile, characterized by an initial elevation of posttraumatic stress symptoms followed by decreasing levels over time. A significant percentage, 45% to 79%, had a resilience profile characterized by low levels of posttraumatic stress symptoms over time [10].

The problem addressed by this component is not reducible to a shortage of educational content or a lack of qualified teachers and volunteers, though both are documented realities of the Ditwah response. The deeper problem is one of system design. Current humanitarian education responses are architecturally misaligned with the neurobiological reality of how trauma affects the learning ready brain. Existing digital education tools deployed in disaster and refugee settings, including Kolibri (Learning Equality), KA Lite, and Remote Area Community Hotspot for Education and Learning (RACHEL), operate on the implicit assumption that the learner is psychologically prepared to receive and process academic content. This assumption, which may be reasonable in stable educational contexts, is neurologically unfounded for acutely traumatized children [11]. Research in trauma neuroscience has established with considerable consistency that sustained activation of the amygdala's threat-detection circuitry during post-traumatic stress suppresses activity in the prefrontal cortex [12], the neural region responsible for working memory, executive function, and the higher-order cognitive processes that academic learning requires. In practical terms, a child in acute post-disaster distress cannot learn mathematics or language, not because the content is inappropriate or the teacher inadequate, but because the neural architecture that would process and retain that content is functionally offline.

Psycho-Attentional Gated Educational System (PAGES) is designed to address this misalignment directly. Rather than providing educational information to all children regardless of their psychological and attentional states, PAGES uses a psycho-attentional gating architecture wherein the combined psychological and attentional states of the child, as measured by passive behavioral interaction signals, determine whether the child is allowed to

engage with cognitive educational information at all. The psychological and attentional states of the child are not an additional process that occurs concurrently with educational engagement; rather, they are a measurable process that must be satisfied before educational engagement is allowed, which is supported by neuroscientific studies on the effect of trauma on cognition [12]. It is noteworthy that PAGES does not implement adaptive learning, wherein educational content is modified according to the performance of the child. Rather than this approach, PAGES uses a gating architecture wherein readiness is measured, a threshold is evaluated, and only upon successful completion of this process is the child allowed to engage with standardized cognitive educational content.

1.3.1 Affected Populations, Magnitude of Impact, and Stakeholder Analysis

A. Primary Target Population: Children in Primary Education (Ages 7-9)

The main population of interest for the PAGES component is children aged 7 to 9 years, which equates to Grades 3 and 4 of primary education in Sri Lanka, as run by the Ministry of Education and the National Institute of Education (NIE). This age range is more than just a statistical construct; it is supported by two lines of evidence that converge to provide a strong rationale for its selection.

Merzenich, along with other recent studies in the field of developmental neuroscience, has identified that the range of 6 to 11 years old represents the peak period of synaptic density and the formation of neural pathways in the human brain [13]. From a cognitive developmental perspective, Piaget's theory of cognitive development places children aged 7-11 in the Concrete Operational Stage a period during which logical thinking, categorization, cause-and-effect reasoning, and structured learning first become neurologically viable [14]. Grade 3 specifically represents the internationally recognized 'Grade 3 Gateway,' the critical checkpoint by which foundational literacy and numeracy must be established if long-term learning outcomes are to be protected [15]. United Nations Educational, Scientific and Cultural Organization (UNESCO) and the World Bank both identify Grades 3 and 4 as the single most consequential intervention point in primary education, noting that children who have not achieved foundational literacy by this stage face a compounding deficit that widens progressively through secondary school and into adult economic outcomes [16][17]. Grade 4 extends the window through the completion of the Concrete Operational Stage's most productive phase, enabling the system to capture children at both the entry point and the consolidation point of this critical developmental window.

From a psychological readiness perspective, Blair & Razza's seminal work clearly indicates that self-regulation and attentional readiness are biological pre-requisites for learning among children aged nine or less, not merely facilitators of learning [18]. This is particularly pertinent to the context of the proposed PAGES deployment scenario, a zero-internet, volunteer-staffed scenario following a disaster situation, because, unlike older children who can maintain their own interest through intrinsic academic motivation, Grades 3 and 4 children require psychological readiness before any potentially beneficial cognitive content is presented to them. The gating model is not only theoretically justified but essential for this age cohort.

B. Magnitude of Impact: Systemic Failures Beyond the Classroom

While children constitute the primary end users of PAGES' educational and psychological support functions, an accurate account of the problem's magnitude requires examination of the broader systemic failures that render conventional educational responses structurally inadequate in the disaster camp context. These failures, documented in both the Ditwah field reports and the academic literature, collectively establish that technological mediation is not a supplementary convenience but a structural necessity.

The Ditwah response indicates that the teacher-to-student ratio is approximately 1:77, based on 308 teachers serving 23,695 students in internal displacement camps [19]. This teacher-to-student ratio makes it operationally infeasible to employ methods that require teacher observation, assessment, and/or differentiation of students. A teacher cannot simultaneously attend to 77 students to ascertain each student's cognitive state, detect signs of psychological distress, and adjust the presentation of the subject matter to the needs of individual students. Accordingly, the gating algorithmic layer provided by the PAGES is not an improvement over the status and it is a replacement for human ability that cannot exist in the context in which it is being used.

The problem is further compounded by the trauma profile of the teaching staff. UNICEF Situation Report No. 8 stated that the total number of teachers and school principals in the affected provinces identified as trauma-affected individuals in need of psychosocial support was estimated at 28,900 [2]. This problem presents an ethical dilemma because human actors in the children's process of healing and moving forward are the same ones who need the same process of healing. It is not only optimistic but impossible for trauma-affected teachers to provide trauma-informed teaching to 77 students without technological support.

The wider global context adds further weight to the significance of this locally focused research. According to UNESCO, over 222 million children in 93 countries require humanitarian support for their educational needs each year, and the vast majority of those are in low and middle-income countries that have limited digital connectivity [20]. The systematic review carried out by Crompton et al. [21], which reviewed 60 studies in 48 countries, explicitly highlighted the need for research in this area as a priority gap in the evidence base. They highlighted the fact that internet-based solutions dominate the evidence base, whereas solutions for low-connectivity environments are vastly underrepresented. The issue of a lack of learning tools that can function in an offline capacity is therefore a priority research need.

C. Stakeholder Consultation: Methodology and Institutional Context

The results of the consultations are reflected in the design rationale of the component in the following table: Table 1.

Table 1: Stakeholders and users consultation summary

Stakeholder	**Role / Position**	**Key Pain Points Identified and observations**	**Implications for PAGES**
Mr. P.S. Priyanath Perera	Divisional Secretary, Kolonnawa	• Data Gathering and Entering	• Validates PAGES's requirement to

	(Administrative authority)	• Immediate post-disaster education is infeasible due to psychological shock • Technical human resource limitations constrain digital deployment • Most of the students stopped their education after grades 8 or 10. Suggest collaborating with them on our project to motivate them. • Some children lack support from their families for education. No better motivation for their life and education. • Children are not in a comfortable home, and they may feel unsafe. Fathers may be at home to protect their properties, and they may feel that they are unsafe without their fathers.	register and track individual children from camp arrival and provide physical support when needed. • Giving hope to children that education is not disrupted due to disaster and giving psychological support when required.
Mr. Dhananjaya	Senior Field Officer, National Disaster Relief Services Centre (NDRSC)	• Inaccurate camp data resource misallocation due to non-camp residents accessing supplies. • Camp duration can be more than 2-4 weeks in the Badulla district, and education support is more required.	• Directly validates PAGES's phase-gated design: psychological stabilization before academic content. • Informs offline-first, low-technical-barrier deployment requirement. • Giving hope to children that education is not disrupted due to

			disaster and giving psychological support when required.
Mrs. K. M. A.N. Bandara	Assistant Zonal Educational Director (School Administration), Homagama	• Research permission granted; school access to Grade 3-4 students confirmed for pilot study	• Pilot study access secured for Design and Validate a Combined Psycho-Attentional Assessment Layer and Develop and Evaluate a Machine Learning Anomaly Detection System for Child Welfare Monitoring evaluation; ethical clearance pathway confirmed
Ms. Amunumulla	Primary Teacher, Sri Rahula Maha Vidyalaya, Mulleriyawa (Community stakeholder-education)	• Schools were closed for more than 2 weeks. • Some students showed poor attendance at school for a period. • Personal trauma affecting facilitation capacity is limited, and no structured guidance system is available.	• Validate the volunteer facilitator dashboard • Integrate child-friendly digital tools for resilience • Education with limited resources. • Consider the psychological level before giving education.
Disaster-Affected Children (Ages 7-9, Grades 3 and 4)	Student	• Psychological shock prevents immediate learning engagement. • Unfamiliar camp environment disrupts routine and sense of safety • Hopeless for the future and felt fear	• Confirms phase-gated design necessity. • Educational and psychological support. • Give physical support if recognized by the system.

			• Help to improve the cognitive level of each.

1.4 Statistical Evidence Base

The following quantitative data, verified through primary humanitarian sources documenting the Cyclone Ditwah response, establish the empirical basis for the research problem:

- 890,000+ children affected across nine provinces in Sri Lanka (UNICEF Sri Lanka Humanitarian Situation Report No. 8, February 2026) [1]
- The total number of primary and secondary schools structurally damaged or unsafe in the affected area was 1,682 (UNICEF Situation Report No. 7, January 2026) [2]
- Between 15-20% of the affected schools sustained structural damage precluding immediate educational use (UN Sri Lanka SitRep #7, 2026) [2]
- 500 schools repurposed as displacement shelters, requiring complete evacuation, cleaning, and physical repair before educational function could be restored (Sri Lanka Red Cross Situation Report No. 8, November 2025) [22]
- 308 teachers serving 23,695 students in documented IDP camp configurations, representing a teacher-student ratio of approximately 1:77 [19]
- 28,900 teachers and school principals identified as trauma-affected and requiring psychosocial support, the very population tasked with delivering children's educational recovery [2]
- Only 36 operational child-friendly spaces served 1,600 children out of a total of almost 900,000 people in need of such facilities, indicating a coverage rate of less than 0.2%. (UNICEF, 2026) [1]
- The total number of operational child-friendly spaces was 45, which served 28,504 children and 11,377 caregivers out of the total number of children in need of such facilities. The coverage rate was less than 4%. (UN SitRep #7, 2026) [2]
- The number of children with special needs who required specially adapted learning material was 900. There was no digital adaptive solution deployed at any point in the response (UNICEF, 2026) [1]
- There was no Artificial Intelligence (AI) or adaptive learning technology deployed in response to Cyclone Ditwah. This is the primary technology used to address research problems. [1][2][23]

The final statistics above are particularly significant as research justification. Despite significant levels of humanitarian funding, the overall educational response to the crisis involving 890,000 children was channeled solely through physical kits, human resources, and static, non-adaptable digital media. This digital deficit, however, should not be regarded as an isolation, given the overall pattern that has been identified in multiple disaster scenarios wherein the creation of intelligent, offline-capable, psychologically aware educational systems has lagged far behind the established humanitarian need.

1.5 Critical Analysis of Existing Solutions and their Limitations

A rigorous assessment of the Cyclone Ditwah educational response reveals that existing humanitarian education interventions, while representing genuine and well-resourced efforts, share a set of structural limitations that collectively constitute the justification for the proposed technological approach.

Table 2: Existing Solutions and Their Limitations

Existing Solution	**What It Provides**	**Structural Limitation**
UNICEF Learning Kits (81,300 distributed post-Ditwah) [1]	• Physical education Materials • Age-grouped content • Individualized pacing	• Non-digital • No psychological state detection • Fixed content regardless of trauma-induced cognitive regression • No individual monitoring
Child-Friendly Spaces -45 CFS serving 28,504 children [2]	• Safe play-based learning • Group psychosocial support	• Dependent on trained human facilitators • Covers <4% of affected children • No individual digital data collection • No welfare alert mechanism
Kolibri (Learning Equality)-deployed in Jordan refugee camps [24]	• Offline Raspberry Pi content delivery • Progress tracking • Curriculum support	• No psychological or attentional pre-assessment • No gating model • No welfare anomaly detection • Static content navigation is not algorithmically personalized
Save the Children Digital Packs [25]	• USB/SD card content for CFS; interactive stories; mindfulness games for group use	• Static and non-adaptive • no individual trajectory tracking
TeamUp Methodology- Save the Children / CCH, Sri Lanka [21]	• Structured play and movement for psychosocial recovery;	• Discrete psychosocial program entirely separate from educational content

	evidence-based for PTSD reduction	• No information about the psychological state is shared with the educational context • Structural fragmentation
WHO mhGAP Offline Application [25]	• Mobile tools for healthcare providers • Child assessment modules • Coping strategies	• Designed for adult professional use, not for direct child interaction • No educational function • Clinical rather than educational-recovery focus
Traditional Adaptive Learning Systems (DreamBox, Knewton) [21]	• Content difficulty adjustment based on academic performance data	• Assumes child is cognitively ready; does not assess psychological or attentional state before learning begins • Not designed for disaster contexts • Internet-dependent
Psychological First Aid Programs- 621 workers trained post-Ditwah [2]	• Psychological stabilization reduces PTSD and anxiety via trained volunteers	• Requires trained human facilitators • Not integrated with educational systems • Cannot continuously monitor individual children across sessions • Not scalable to 890,000 children

1.6 Solution Overview

PAGES made five integrated original contributions:

1. A Combined Psycho-Attentional Assessment Layer is to quantify the psycho-attentional readiness of each child using three indirect behavioral measures, thus avoiding direct self-reporting by the child. Specifically, the psycho-attentional readiness score is composed of the following components:
   a) A gamified Go/No-Go paradigm task [26] (50% weight): In this task, the child responds to the target animal by tapping on the screen, avoiding responses to the no-go stimuli, over a 90-second interval. The accuracy, false tap rate, reaction time, and reaction time variance of the responses are recorded.
   b) Passive behavioral interaction signals [27] (30% weight): These comprise the idle time ratio, rapid click count, time to first interaction, and session completion ratio.
   c) A pre-computed rolling baseline score stored within the local profile of the child, which is updated using a weighted rolling average after each session (New Baseline = Old Baseline × 0.8 + Today's Score × 0.2) (20% weight).

2. A Psycho-Attentional Gating Model that determines, based on the composite score against a defined threshold, whether a child is in a neurologically viable state to engage with cognitive learning content. Children below threshold receive a brief calming or re-engagement support activity and are reassessed before any cognitive content is offered. Children above the threshold proceed directly to the cognitive layer. This gating architecture is grounded in Bloom's Mastery Learning Model [29], Vygotsky's Zone of Proximal Development (ZPD) [30], Blair and Razza's neuropsychological evidence [18], and Van der Kolk's trauma neuroscience [11].

3. A Standardized Cognitive Content Layer delivering identical, curriculum-aligned cognitive activities to all children who pass the psycho-attentional gate. Content standardization at the delivery level, combined with individualized gating at the readiness level, enables valid cross-child performance comparison while respecting individual psychological differences consistent with Bloom's conditional standardization principle [29] and directly serving the NIE Sri Lanka Grade 3-4 curriculum framework [31].

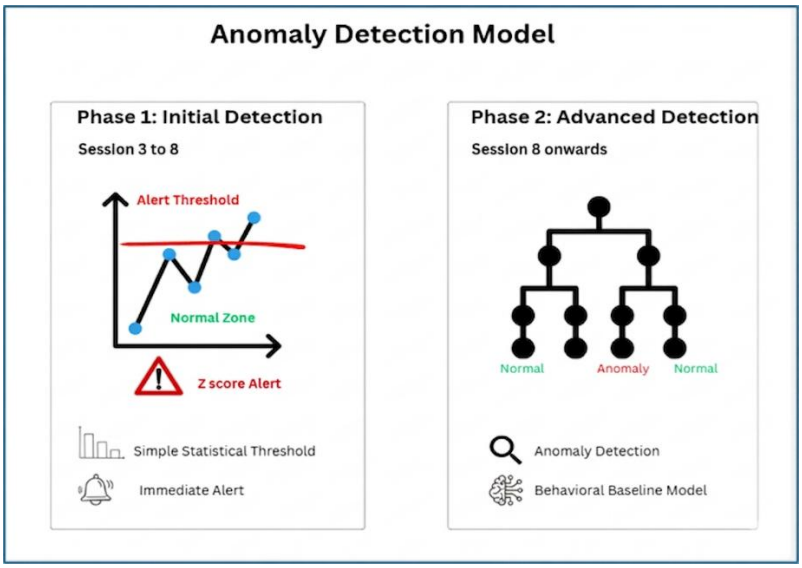
4. 

Figure 1: Anomaly Detection Model

A Machine Learning Anomaly Detection System (Figure 1) employing a Z-Score statistical method (Phase 1, operational from session three per child) progressing to an Isolation Forest model[32] (Phase 2, after sufficient session accumulation) that identifies children exhibiting persistent psycho-attentional distress across multiple sessions specifically three or more consecutive flagged sessions, or three of five recent sessions and generates plain-language, actionable welfare alerts for administrators via the synchronized cloud dashboard.

5. An Offline-First IoT Deployment Architecture in which the web application, local SQLite database, ML inference, and gating logic operate entirely on the Smart-Camp Box without internet dependency. A Sync Model automatically uploaded session data, child profiles, and anomaly flags to cloud storage upon WiFi availability, enabling admin dashboard access, cloud-based model improvement, and updated model redeployment to the device.

2. LITERATURE REVIEW

2.1 Trauma Neuroscience and the Neurobiological Basis of Learning Disruption

2.1.1 Van der Kolk - Trauma's Impact on the Learning Brain

The work by Van der Kolk, “The Body Keeps the Score” (2014) [11], constitutes the most comprehensively evidenced account of the neurobiological impact on children's cognitive function currently available in the literature. By integrating the results of decades-long research into the clinical and neuroimaging consequences of trauma, Van der Kolk shows that the sustained hyperactivation of the amygdala during post-traumatic stress results in the suppression of prefrontal cortex activity. This is the primary neurobiological rationale for the PAGES gating model. This is because, before any cognitive content is provided, the PAGES model must first ensure the psycho-attentional stability of the individual. This is because not only is the presentation of any academic content to the traumatized individual irrelevant, but it can also re-traumatize the individual by associating the educational environment with the neurologically compromised state of failure. The primary limitation of Van der Kolk's work, as it pertains to the PAGES model, is the fact that the research is based almost exclusively on high-income populations and does not directly address the potential of technology-based behavioral signals to detect attenuated variants of these stress states. This is the purpose of the evaluation component of the current work.

2.1.2 Blair and Razza -Readiness as Biological Prerequisite

Blair and Razza's landmark study on effortful control and executive function in young children [18] established that psychological self-regulation functions as a prerequisite biological condition for learning, not merely a facilitating factor. Children who lack emotional and attentional self-regulation demonstrate significantly lower emergent literacy and numeracy skills, independent of curriculum quality or instruction time. This directly justifies the gating architecture of PAGES: confirming psychological and attentional readiness before cognitive engagement is not a pedagogical preference but a neurological necessity for Grades 3-4 children. While some affective computing systems attempt to detect stress or disengagement, PAGES is novel in explicitly operationalizing this principle as a computational content gate.

2.2 Educational Psychology Frameworks Underpinning the Gating Architecture

2.2.1 Maslow's Hierarchy of Needs

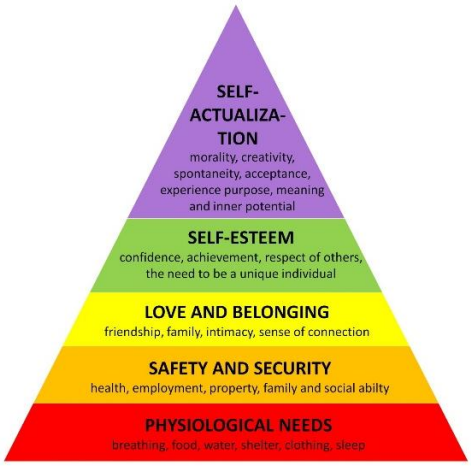


Figure 2: Maslow's Hierarchy of Needs

Maslow's motivational hierarchy [33] (Figure 2) provides the macro-level theoretical justification for PAGES's gating model. The hierarchy's central thesis is that higher-order cognitive and developmental needs cannot be adequately addressed until lower-order safety and belonging needs are met. PAGES operationalizes Maslow's principle of sequential prioritization of needs: a child does not receive cognitive content until behavioral signal data confirms that their psycho-attentional state has achieved a threshold level of stability. Academic content is delivered only when behavioral signals confirm sufficient psycho-attentional stability. The Cyclone Ditwah field documentation supports this framing, with UN SitRep data confirming that Maslow-informed sequential response protocols, physiological needs addressed first, psychological safety second, and academic education third, were identified as most effective across comparable disaster contexts, including Cyclone Idai, Mozambique, 2019 [34]. PAGES is, to the best of the authors' knowledge, a proposed computational implementation of Maslow's hierarchy as a dynamic content-access gate in an IoT-deployed educational technology system.

2.2.2 Vygotsky's Zone of Proximal Development

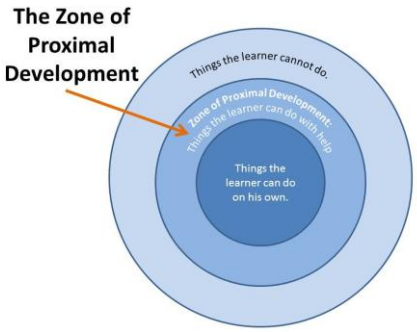


Figure 3: Vygotsky's Zone of Proximal Development

Vygotsky's Zone of Proximal Development [30] (Figure 3) defines the optimal learning zone as the space between independent performance capacity and the upper boundary of guided performance capacity. In the PAGES context, the ZPD is not used to calibrate content difficulty; it is standardized, not adapted, but rather to define the precondition for ZPD-aligned learning to be possible at all. A child in acute psychological distress is, neurologically, outside any productive ZPD: trauma-suppressed prefrontal function means that even appropriately levelled content cannot be processed. PAGES's gate operationalizes the ZPD's implicit precondition: a child must be in a neurologically viable state before the concept of 'just-right challenge' has any meaning. The Perplexity research synthesis (2026) [35] confirms that ZPD-aligned offline tools could extend kit-based learning in Sri Lankan camps. PAGES provides the psycho-attentional prerequisite infrastructure that makes such ZPD alignment educationally productive.

2.2.3 Bloom's Mastery Learning and Conditional Standardization

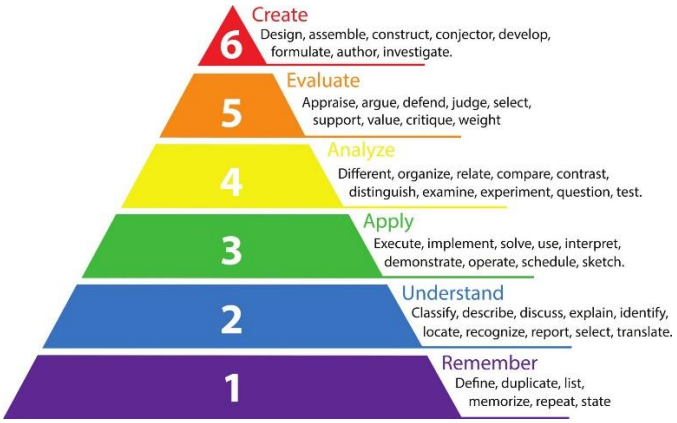


Figure 4: Bloom's Taxonomy of Learning Domain Levels

Bloom's Mastery Learning Model (1968) [29] (Figure 4) demonstrated that when students are brought to a sufficient readiness level before instruction, standardized content produces strong and equitable learning outcomes across diverse learners. This provides the theoretical basis for PAGES's standardized cognitive content layer: content uniformity is not a simplification, but a methodologically deliberate choice supported by Bloom's evidence, enabling meaningful cross-child performance comparison. The distinction between PAGES and adaptive learning systems is sharpened by Bloom's model: PAGES adapts the gate (individualized readiness assessment) rather than the content (standardized for all who pass). This is a theoretically coherent and evidence-grounded architectural choice, not a constraint.

2.3 Behavioral Assessment Methods for Children

2.3.1 Conners' Continuous Performance Test and Go/No-Go Paradigms

Conners' Continuous Performance Test (CPT, 1994) [27] is a widely used computerized measure of sustained attention in children. It was established that attention and psychological

arousal in children can be reliably measured through behavioral response patterns, reaction time, error rate, and consistency without self-report. The CPT has been validated for children as young as six years old. Cragg and Nation (2008) [26] specifically studied Go/No-Go response inhibition tasks with children aged 6-11, confirming that response accuracy and reaction time in simple tap/don't-tap games are valid, age-appropriate measures of attentional control and psychological arousal in this population. The PAGES simple tasks are a child-friendly touchscreen implementation of this paradigm, yielding the same four measurement dimensions (accuracy, false responses, reaction time, and reaction time variance) in a format appropriate for Grade 3 children with no self-report requirement. Sergeant et al. (2000) [36] further confirmed that reaction time variability is the single most sensitive indicator of attentional state in children, a finding directly incorporated in the PAGES composite scoring algorithm.

2.3.2 D'Mello and Graesser - Multi-Signal Behavioral Fusion

D'Mello and Graesser (2012) [37] demonstrated that combining multiple behavioral signals produces significantly more accurate learner state detection than any single signal in isolation. Their multi-signal fusion approach to interacting data provides the methodological precedent for the PAGES three-source fusion model (50:30:20 weighting of gamified task, passive signals, and historical baseline). The 50% weight assigned to the gamified task reflects its status as the freshest, most direct behavioral measurement from the current session; the 30% weight for passive signals reflects continuous in-session behavioral monitoring; and the 20% historical baseline weight provides contextual stability against environmental noise, a critical consideration in the chaotic conditions of a disaster relief camp.

2.4 The Gating Model vs. Adaptive Learning

2.4.1 Corbett and Anderson - Knowledge Tracing and Its Limitations

Corbett and Anderson's (1994) Knowledge Tracing model [21], the foundational architecture for adaptive tutoring systems, adjusts content difficulty based on observable performance data, including correct and incorrect responses. Modern adaptive learning systems, including Deep Knowledge Tracing (Piech et al., 2015) and commercial platforms including DreamBox and Knewton, operate within this performance-based adaptation paradigm. The critical limitation of this paradigm in the post-disaster context is its universal assumption that the learner is cognitively ready to engage and that performance variability reflects differences in domain knowledge rather than psychological or attentional state. In disaster-affected Grade 3-4 children, this assumption is specifically and systematically violated: poor performance is as likely to reflect trauma-induced cognitive suppression as genuine academic knowledge gaps. PAGES addresses this by introducing a pre-learning state assessment and gating layer that performance-based adaptive systems largely omit, not by adapting content difficulty, but by confirming readiness before any content is delivered.

2.5 Offline IoT-Based Educational Technology

2.5.1 Kolibri and Raspberry Pi Deployment - Precedent and Limitations

Kolibri, documented by the Organization for Economic Co-operation and Development (OECD) (2021) [24] for deployment in Jordan refugee camps using Raspberry Pi servers, provides a directly relevant existing system to PAGES's deployment architecture. Kolibri's widespread use in over 200 countries demonstrates the feasibility of the Raspberry Pi offline server model for humanitarian contexts. However, the OECD evaluation explicitly noted that Kolibri's absence of adaptive personalization limited its effectiveness for heterogeneous camp populations. PAGES extends significantly beyond Kolibri by adding a psycho-attentional assessment layer, a gating model, anomaly detection for welfare monitoring, and automatic cloud synchronization, transforming a static offline content library model into a psychologically aware, welfare-monitoring educational system.

2.6 Machine Learning Anomaly Detection for Child Welfare Monitoring

2.6.1 Isolation Forest Unsupervised Per-Child Anomaly Detection

Liu et al.'s (2008) Isolation Forest algorithm [36] is the proposed Phase 2 anomaly detection model for PAGES, as shown in Figure 1. Isolation Forest operates unsupervised, requiring no pre-labeled 'abnormal' examples, making it ideal for the PAGES context where labeled instances of child psychological distress patterns are unavailable. It isolates anomalies through random feature partitioning: anomalies, being statistically rare and behaviorally different from the child's personal baseline, require fewer partitions to isolate than normal observations. Chandola et al.'s (2009) comprehensive survey [38] confirms that Isolation Forest-type approaches are most appropriate when individual behavioral baselines vary significantly across subjects and labeled anomalies are scarce, exactly the PAGES deployment condition. The two-phase architecture Z-Score statistical detection from session three onwards (Phase 1) transitioning to Isolation Forest after sufficient data accumulation (Phase 2) mirrors common practice in patient healthcare monitoring, where simple statistical thresholds provide immediate alerting while more sophisticated models are trained on accumulating patient data [38]. This phased approach ensures PAGES is functional from the first sessions while progressively improving accuracy over time.

2.7 Post-Disaster Education: Systematic Review Evidence

Crompton et al.'s (2021) systematic review [21], examining 60 studies across 48 countries between 2010 and 2020, represents one of the most comprehensive systematic reviews to date on educational technology in crisis contexts. The review found that while technology-assisted education in emergencies produced generally positive engagement outcomes, adaptive platforms were significantly underrepresented in the evidence base, and internet-based tools systematically excluded the most vulnerable learner populations in low-

connectivity settings. The review highlights the lack of offline-capable adaptive tools as a priority research gap, a gap that PAGES's gating model and offline-first IoT architecture directly addresses. The review's finding that 'personalized digital learning produced engagement gains even in low-resource settings' is consistent with the PAGES approach, while the caveat that tools requiring internet connectivity 'systematically excluded the most vulnerable' directly motivates the offline-first architectural commitment.

2.8 Research Gap

The literature review section identifies a research gap that is both specific and significant.

2.8.1 No Pre-Learning Psycho-Attentional Gating in Any Existing Educational Technology

All existing educational technology systems, from Knowledge Tracing [21] to commercial-oriented adaptive systems, share the fundamental assumption that the student is ready to learn and that the adaptation of the content is a reaction to the student's performance indicators after the learning. Currently, no educational technology system assesses, before the start of the learning process, whether the child is psychologically and attentively ready to learn any content at all. As shown by the neuroscientific evidence presented in Sections 1.1.1 and 2.1 above, the assumption is explicitly false for disaster-affected children between the ages of 7 and 9, where psychological distress physiologically inhibits the cognitive capacity for learning [11][3]. An educational content delivery system for a psychologically distressed child is not only useless; it is also a form of pedagogical harm because it requires the cognitive performance of a child who is neurologically incapable of delivering it. This paper presents the first documented proposal for the operationalization of the readiness gates before the learning process for a deployable educational technology system for young disaster-affected children.

2.8.2 No Behaviorally Indirect Psycho-Attentional Assessment for Non-Self-Reporting Children in Educational Technology

Existing educational technology systems for learner assessment mainly focus on facial expression-based assessment using webcams, physiological sensors, and direct self-report questionnaires [37], which are impossible for Grade 3-4 children in a disaster camp environment. Using webcams and physiological sensors requires costly hardware that cannot be utilized with the Smart-Camp Box. Facial expression-based assessment using images and videos of children poses critical issues regarding child data privacy. Direct self-reporting is not suitable for children below the age of 9, who lack introspection to accurately report their own states of mind [28]. The PAGES method, using a gamified Go/No-Go behavioral task [26], passive interaction signals [27], and a pre-computed rolling historical baseline, represents a methodologically unique and hardware-minimal solution for psycho-attentional state assessment in young non-self-reporting children. This unique combination of assessment sources for this specific learner population is not found in any other published system.

2.8.3 No Anomaly Detection for Child Welfare Monitoring in Educational IoT Systems

Existing educational IoT and EdTech systems invariably produce learner analytics focused only on academic performance metrics such as assessment scores, time spent on tasks, topic mastery, and learning curves. Currently, no such system uses anomaly detection techniques to detect children who display patterns of psychological distress across sessions and produce actionable welfare notifications for physical intervention. This is especially relevant in disaster scenarios. It is documented by Save the Children that tens of thousands of children in temporary shelters need mental health support [25], with a significant lack of qualified counsellors. The anomaly detection system developed for the PAGES project is especially relevant for such a disaster scenario, where the anomaly detection system enables non-specialist admin who do not have psychological training to receive notifications about children who display patterns of psychological distress based on the data collected from the children. This human-in-the-loop welfare monitoring architecture has no precedent in existing educational technology literature.

2.8.4 No Offline-First IoT System Combining Psycho-Attentional Gating and Welfare Monitoring for Post-Disaster Deployment

Although the technical feasibility of offline content delivery for the Raspberry Pi-based system was shown in the context of the Kolibri offline content delivery system [24], and despite the presence of IoT-based educational systems for well-resourced contexts, no system currently exists that integrates all of the following within a single, deployable component for use in post-disaster contexts: (a) an offline-based IoT system, (b) pre-learning assessment of psycho-attentional states through behaviorally indirect means, (c) control of cognitive content access based on learner readiness, (d) machine learning-based anomaly detection for welfare, and (e) automatic cloud synchronization for the persistence of data and model updates. This was explicitly identified as a research priority for the educational technology community, as noted by Crompton et al. (2021) [21]. In fact, the Rapid Evidence Review conducted by the EdTech Hub found that offline adaptive tools can help to mitigate infrastructure and equity barriers for crisis contexts, though it found that the overall evidence for such tools was scarce. Accordingly, the PAGES system addresses the research priority.

2.9 Why Current Approaches Are Insufficient

2.9.1 Technological Insufficiency

The educational platforms that are currently available are designed to work with consistent connectivity, hardware availability, and educated instructors. They are not designed to run on a portable IoT device with low connectivity, operated by a admin with no technical expertise, and accessed by children on shared devices in an overcrowded disaster camp. The offline-focused, IoT-based deployment model of PAGES fills the technological gap.

2.9.2 Architectural Insufficiency

Existing adaptive systems change the content of instruction based on a child's performance. PAGES, however, poses a fundamental, architecturally prior question: is instruction appropriate at a given time for a particular child? This is a paradigm shift, not a variation of existing adaptive learning systems, and is supported by a body of research in developmental psychology and trauma neuroscience, which existing systems have neither attempted nor validated. The lack of a gating layer is not merely a problem to be solved but a fundamental architectural choice based on the belief that a learner is always ready to learn. PAGES systematically challenge and computationally answer this assumption.

2.9.3 Humanitarian Insufficiency

Existing systems provide learner analytics to trained educators in a predictable institutional environment. In contrast, PAGES provides welfare alerts to admins in chaotic disaster environments, where a child's psychological well-being may have greater immediate implications for their welfare than their academic achievement. There is no existing educational technology system with this humanitarian priority structure, in which monitoring child welfare is an equal to or greater priority than providing educational content.

2.10 Justification of Novelty and Research Contribution

The application of pre-learning psycho-attentional gating based on Piaget's stages of cognitive development [5], Maslow's hierarchy of needs [33], Bloom's mastery learning theory [29], Vygotsky's Zone of Proximal Development [30], and the neuroscientific studies by Van der Kolk, Blair, and Razza [11][3], and packaged into a deployable form of educational technology, is a theoretically novel contribution to the fields of Information Technology Services (ITS) and affective computing.

The novel approach to behavioral fusion from three sources (gamified Go/No-Go task, passive interaction signals, and rolling historical baseline) for psycho-attentional measurement is a methodologically novel approach for the measurement and assessment of psycho-attentional states without the need for direct self-reporting from children, with no direct precedent in published research on educational technology.

The offline Internet of Things (IoT) architecture combining machine learning scoring, behavioral measurement, Isolation Forest anomaly detection, and automatic cloud synchronization into a single portable device is a technically novel approach to the deployment of Internet of Things (IoT) architecture for the deployment of AI in education.

The post-disaster primary education context in Sri Lanka, supported by empirical data from Cyclone Ditwah referenced by [1][2], is a contextually novel and underserved context for the deployment of AI in education, which existing published research on educational technology has not satisfactorily addressed. This is a research gap identified by Crompton et al. [21].

2.11 Evidence-Based Argument for Research Necessity

Empirical Scale: The Cyclone Ditwah (2025) disaster affected 890,000 children, destroyed 1,682 schools, resulted in a teacher-to-student ratio of 1:77, and resulted in tens of thousands of children in relief camps who require both educational support and mental health care, with zero AI-driven educational systems in place to support the response efforts [1][2].

Neuropsychological: The psychological distress of children in the 7-9 year age group neurologically inhibits the prefrontal cognitive functions necessary for learning, making it both ineffective and counterproductive to educate children in a distressed psychological state [3][18].

Methodological: The behavioral indirect methods of Go/No-Go Gamified Tasks and Passive Interaction Analysis offer a reliable and valid method of measurement of the psychological and attentional state of children in the 6 to 11 year age group, without the requirement of any specialized hardware or self-report, and have been supported through several peer-reviewed studies [26][27][37].

Technical: The technical viability of an offline-first IoT-based educational system using Raspberry Pi hardware in a post-disaster setting, while educationally effective in settings such as the Jordanian Refugee Camps with Kolibri, does not exist when incorporating a psycho-attentional gating mechanism or welfare anomaly detection in a low-connectivity network.

Literature Gap: No current educational technology exists that incorporates a psycho-attentional pre-learning gating mechanism, behavioral state measurement, machine learning-based anomaly detection for welfare monitoring, and IoT deployment in a single component for post-disaster primary education support, low connectivity, as evidenced in the literature [21][38].

3. METHODOLOGY

3.1 Research Approach

3.1.1 Research Paradigm

This research adopts a Design Science Research (DSR) paradigm as its primary methodological framework. DSR is appropriate when the goal is to design, build, and evaluate an artefact that solves a real-world problem which precisely describes the PAGES component. The artefact in this study is the PAGES system itself: a working software system that integrates psycho-attentional assessment, gating logic, standardized cognitive content, machine learning anomaly detection, and offline IoT deployment. Following *Hevner et al*. (2004), the research proceeds through three design science cycles: the relevance cycle (grounding the design in the real-world problem of post-disaster primary education), the design cycle (iterative design, build, and evaluation of PAGES), and the rigour cycle (grounding design decisions in existing knowledge from trauma neuroscience, behavioral assessment, and anomaly detection literature).

Within the DSR framework, the study incorporates an experimental evaluation component for pilot validation. A within-subject quasi-experimental design is used during the pilot phase, in which each child serves as their own control across session conditions, enabling valid

comparison of system behavior under varying attentional states without requiring a separate control group.

3.1.2 Research Design Overview

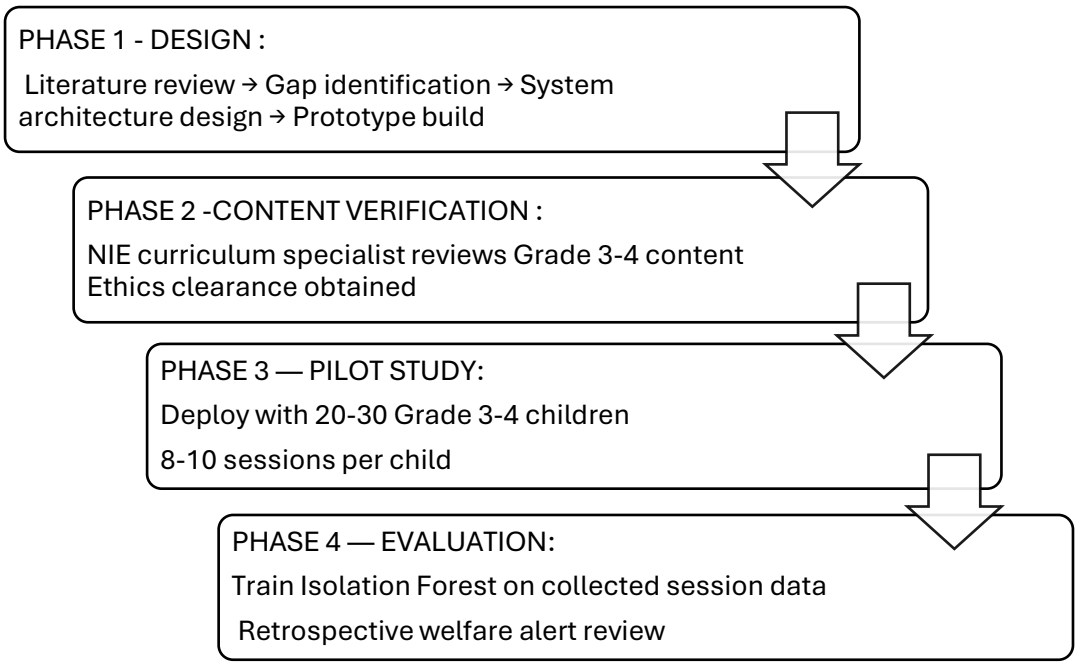


Figure 5: Research Design Overview

3.1.3 Rationale for Participant Population and Context

Due to the temporal distance from the acute phase of Cyclone Ditwah (approximately four months at the time of the pilot study), acutely displaced children in active relief camps were not available to participate.Evaluation of PAGES under conditions of genuine acute psychological distress is identified as a priority for future research.

3.2 Technical Methods and Algorithms

3.2.1 Combined Psycho-Attentional Assessment

The psycho-attentional assessment layer fuses three behavioral data sources into a single composite score (0-100) using a weighted fusion formula:

Composite Score = (Gamified Task Score × 0.50) + (Passive Signal Score × 0.30) + (Rolling Baseline × 0.20)

Source 1 - Gamified Go/No-Go Task (50% weight)

A 90-second touchscreen task based on the Go/No-Go inhibitory control paradigm validated by Cragg and Nation (2008) [26] for children aged 6-11. The child taps target animals (Go

stimuli) and suppresses responses to distractor animals (No-Go stimuli). Four metrics extracted are in Table 3.

Table 3: Gamified metrics

Metric	**Formula**	**What It Measures**
Accuracy	RateCorrect taps ÷ Total target appearances	Sustained attention and task engagement
False Tap Rate	Distractor taps ÷ Total distractors	Impulse control is elevated under distress
Average Reaction Time (ms)	Mean of valid tap response times	Cognitive processing speed
Reaction Time Variance	Standard deviation of reaction times	Attentional consistency most sensitive indicator (Sergeant et al. 2000)

Source 2 - Passive Behavioral Interaction Signals (30% weight)

Four signals (Table 4) collected continuously from the child's browser session without any additional input:

Table 4: Passive behavioral interaction signals

Signal	**Definition**	**Distress Indicator**
Idle Time Ratio	Seconds with no interaction ÷ Session duration	High idle = disengagement
Rapid Click Count	Taps above the 3 per second threshold	Elevated = anxiety or frustration
Time to First Interaction (ms)	Duration from page load to first tap	Long delay = hesitancy or withdrawal
Session Completion Ratio	Screens completed ÷ Total screens	Low = inability to persist

Source 3 - Rolling Historical Baseline (20% weight)

A pre-computed baseline score for each child, updated after every session:

0.8/0.2 weighting applies a smoothing decay that gives more weight to the child's established history than to any single session, preventing one anomalous session from distorting the baseline.

$$\text{New Baseline} = (\text{Previous Baseline} \times 0.8) + (\text{Current Score} \times 0.2)$$

3.2.2 Psycho-Attentional Gating Algorithm

The gating model applies a binary threshold decision per session per child:

```
If Composite Score ≥ Threshold:
        GATE = PASS
                deliver cognitive content
If Composite Score < Threshold:
        GATE = FAIL
                delivering calm support activity
                 reassess after activity
                 second gate decision
                 If FAIL: mark session as FAIL_CONFIRMED,
                 increment consecutive flag counter
```

The readiness threshold is initialized at 55/100 based on the midpoint of the moderate readiness range and is subject to calibration during pilot evaluation. The threshold is applied uniformly; it does not adjust per child. What adjusts per child is the rolling baseline that forms 20% of the composite score, providing individualization at the input layer rather than the decision layer.

3.2.3 Machine Learning Anomaly Detection - Two-Phase Architecture

Phase 1 - Gating and Z-Score Statistical Method (Session 3 onwards)

The Z-Score method requires no training data and activates as soon as three sessions exist for a child:

Z = (Current Composite Score − Child's Mean Score) ÷ Child's Standard Deviation

Threshold: Z < -2.0 → session flagged

Interpretation:

Z = -1.0 → slightly below average (no flag)

Z = -2.0 → bottom 2.3% of normal distribution (flagged)

Z = -3.0 → extreme deviation (flagged, high concern)

The child's mean and standard deviation are recalculated after each session using the entire session history, making the method increasingly accurate as sessions accumulate. If the gate decision is FAIL_FIRST or FAIL_CONFIRMED then the session is flagged as an anomaly. This approach makes sure that individual anomaly detection is accurate.

Phase 2 - Isolation Forest (Session 8 onwards)

Isolation Forest (Liu et al. 2008) [32] is an unsupervised anomaly detection algorithm that identifies anomalies by measuring how easily a data point can be isolated from the rest using random decision trees. Anomalous sessions, being rare and behaviorally distinct from the child's normal pattern, require fewer splits to isolate and therefore receive lower anomaly scores.

3.2.4 Rolling Baseline Update Algorithm

```
def update_baseline (old_baseline, current_score, alpha=0.8):
        Exponentially weighted moving average.
        alpha = weight given to historical baseline.
        (1 - alpha) = weight given to current session.
 new_baseline = (old_baseline * alpha) + \ (current_score * (1 - alpha))
return round (new_baseline, 2)
```

The alpha value of 0.8 was selected to provide a stable baseline that responds gradually to genuine improvement while remaining resistant to single-session noise, consistent with the exponential smoothing literature for short time-series data.

3.3 Data Collection and Ethical Considerations

3.3.1 Participant Profile

Table 5: Participant Profile

Criterion	Specification
Age	8-9 years (Grade 3 and Grade 4)
Location	Mullegama Maha Vidyalaya and Jalthara Maha Vidyalaya
Background	Children affected by or recovering from Cyclone Ditwah (November 2025) |
Number	30 children who are affected by Ditwah
Consent	Written parental/guardian consent is required before participation
Voluntariness	Participation entirely voluntary; withdrawal at any time without consequence

3.3.2 ZEO Permission and Content Verification Protocol

The Zonal Education Division, Homagama, granted permission for the pilot study. The ZEO has set the following condition before children interact with the system: The cognitive content of PAGES must be reviewed and verified as appropriate for Grade 3 and Grade 4 students. Verification process:

- Step 1: PAGES cognitive content layer built (literacy, numeracy, comprehension activities)
- Step 2: Content submitted to ZEO-designated reviewer for alignment review against NIE Sri Lanka Grade 3-4 curriculum framework
- Step 3: Reviewer confirms content is appropriate and provides written sign-off
- Step 4: ZEO grants final permission for child participant engagement
- Step 5: Parental consent forms distributed through the school
- Step 6: Pilot study begins

3.3.3 Data Collection Procedure

Each session was conducted in a supervised classroom setting. A facilitating teacher is present throughout each session block. Children access the PAGES application through a tablet browser connected to the Smart-Camp Box WiFi hotspot - no application installation is required. Sessions are conducted in groups of 5-8 children, each on an individual device.

- Session Structure (15-20 minutes):
    1. Teacher introduces the session (2 min)
    2. Child opens browser, enters child ID code (1 min)
    3. Gamified Go/No-Go task - Tap the Animal (90 sec)

4. Gate decision stored (automatic, < 5 sec)
5. Session ends - data saved automatically

3.3.4 Ethical Considerations

University Ethics Clearance

A full ethics application was submitted to the University Research Ethics Committee. The application included a research summary, participant information, and what data will be gathered.

Child Data Protection

At no point is the actual child's name stored in the PAGES system. Identification is achieved through anonymous codes (for example, HMG-001), which are used at registration. The relationship between the actual identities and the anonymous codes is maintained in paper form only by the class teacher and not in any digital format. Session data is stored locally on the Raspberry Pi with password protection and is transmitted to cloud storage using the MQTT protocol.

Minimizing Psychological Risk

The gamified task has been designed to focus on the aspect of fun rather than difficulty, and the Go/No-Go paradigm is non-stressful for children between 6-11 years of age, according to Cragg & Nation (2008) [26]. The soothing activity provided upon the failure of the gate has been designed to be non-corrective and does not convey any negative message to the child. Children who are constantly failing the gate are not penalized but are provided with a soothing activity and are still positively engaged by the system.

Right to Withdraw

Any child may stop at any time without any consequence. The class teacher is empowered to remove any child from a session at their discretion. Parents may withdraw their child from the study at any time and request deletion of their child's data.

3.4 Validation Strategy and Performance Metrics

3.4.1 Overview

Validation is structured around the five Specific Objectives defined in Section 2.2. Each objective has defined measurability targets that constitute pass/fail criteria for the evaluation.

3.5.2 Assessment Layer Validation

Validate: Does the composite psycho-attentional score reliably detect differences in attentional state for the same child?

Table 6: Assessment layer validation

Metric	Target	Method
Within-child score sensitivity	Mean stressors score significantly lower than baseline ($p < 0.05$)	Paired t-test
Gamified task consistency	Test-retest reliability $\alpha \geq 0.70$ across baseline sessions	Cronbach's alpha
Composite score range utilization	Scores span at least 30 points across all sessions	Descriptive statistics

Justification for revised approach: The original target of $r \geq 0.65$ correlation with volunteer ratings was revised to a within-subject experimental comparison following recognition that per-session volunteer rating at scale is not practically feasible with the available facilitator capacity. The within-subject design provides equivalent statistical validity for detecting assessment sensitivity, as it controls all between-child confounds by using each child as their own control.

3.5.3 Gating Model Validation

Validate: Does the gate correctly distinguish between ready and less-ready states, and does the support activity help children recover?

Table 7: Gating model validation summary

Metric	Target	Method
Gate pass rate differential	Significantly higher pass rate in baseline vs stressor sessions ($p < 0.05$)	Chi-square test
False positive rate	$\leq 15\%$ of stressor sessions incorrectly receive Pass	Condition-tagged session analysis
Session recovery rate	$\geq 60\%$ of Fail-first sessions pass on the second attempt	Gate decision log analysis

3.5.4 Cognitive Content Layer Validation

Validate: Is the content curriculum aligned, and do children improve over sessions?

Table 8: Cognitive content layer validation summary

Metric	Target	Method
Curriculum alignment	≥ 90% of activities rated as aligned by the ZEO reviewer	Content review checklist
Performance progression	Statistically significant improvement across a minimum of 5 sessions ($p < 0.05$)	Linear regression on session scores per child
Session completion rate	≥ 80% of gate-passing children complete the full content session	Session completion ratio from SQLite

3.5.5 Anomaly Detection Validation

Validate: Does the system correctly identify children showing persistent attention distress and generate timely welfare alerts?

Table 9: Anomaly detection validation summary

Metric	Target	Method
Alert precision	≥ 0.75 (75% of alerts confirmed as warranted)	Retrospective teacher review
Alert recall	≥ 0.70 (70% of struggling children flagged)	Teacher-identified missed cases
Alert latency	≤ 3 sessions from pattern onset to alert generation	Session log timestamp analysis
Z-score vs. if agreement	≥ 80% agreement on flagged sessions where both phases are active	Cross-method flag comparison

3.5.6 IoT Architecture Validation

Validate**:** Does the complete system work offline under realistic deployment conditions?

Method: Technical system testing.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 System Diagram of Components

Figure 5 illustrates the high-level system architecture of the PAGES system, showing the three architectural layers, the five key edge-layer components, the data flows between components, and the boundary between offline-first edge-layer operations and cloud-based operations. The system is designed with multiple layers so that all consumer-facing and safety-

critical operations occur within the edge layer, even without connectivity. In contrast, cloud-based operations provide enhancements and oversight when connectivity is available.

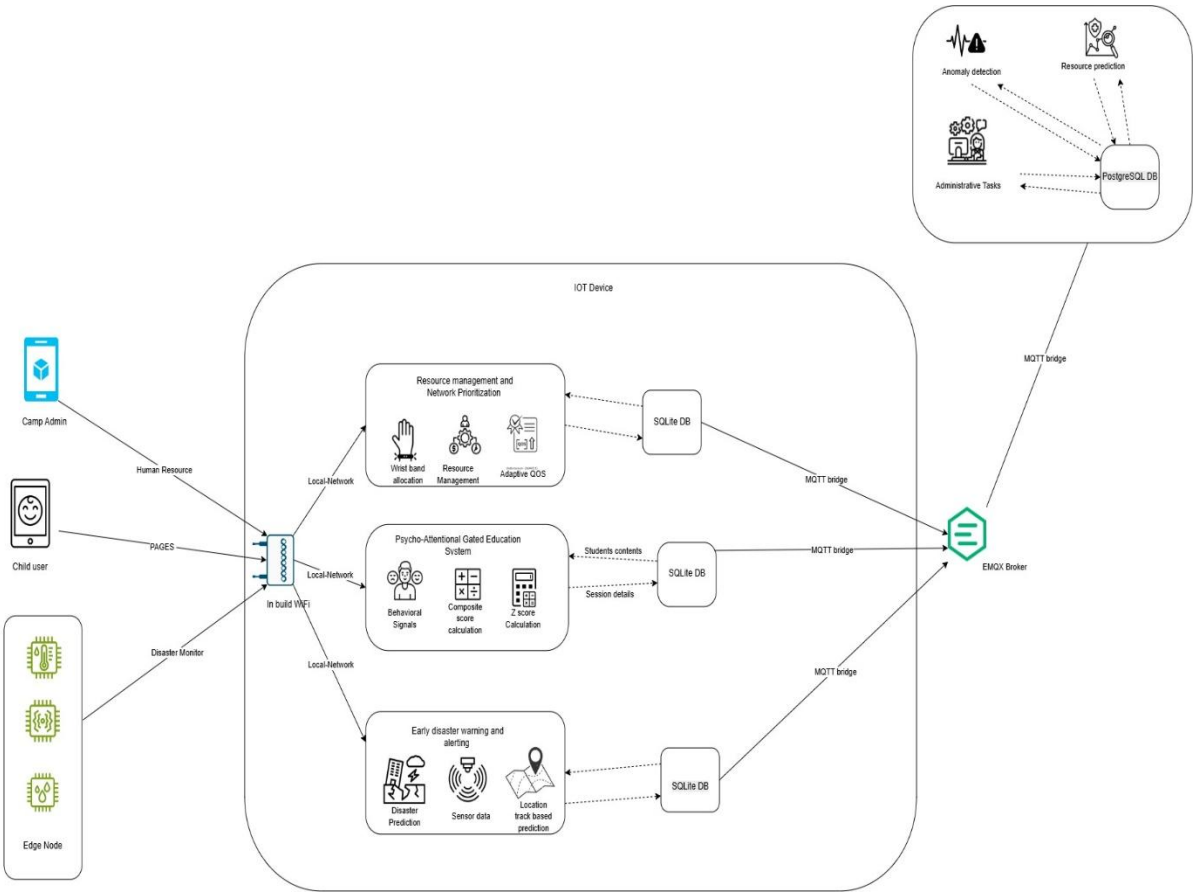


Figure 6: PAGES architectural diagram

4.2 Session Pipeline Flowchart

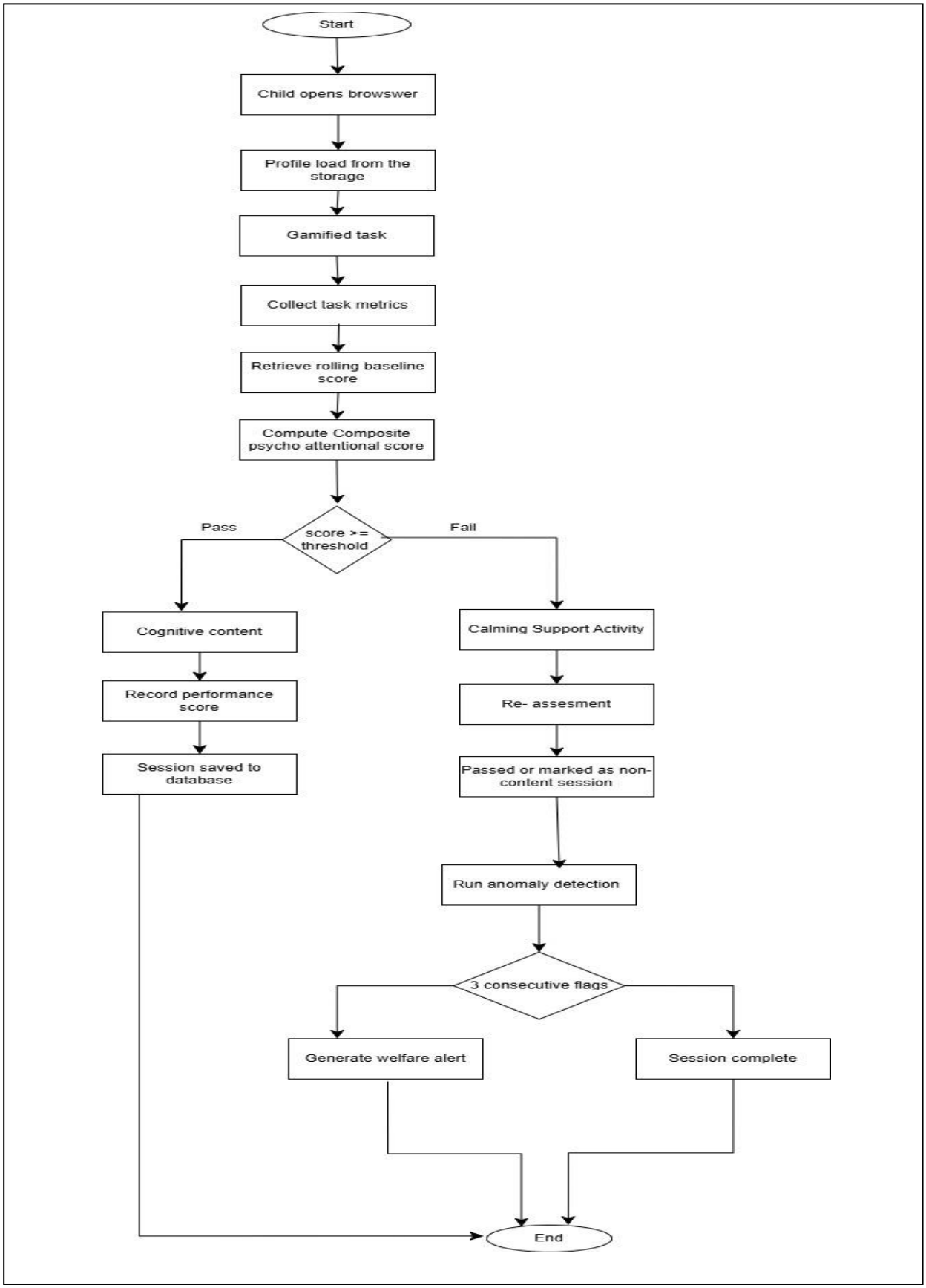


Figure 7: Session flowchart

4.3 Data Flow Explanation

Table 10: Data flow summary

Task	Source	Destination	Data content	Condition
Child interaction data	Child device (browser)	PAGES App (wlan0)	Tap events, timestamps, and accuracy signals from the gamified task	During every session, a continuous stream
Composite score	Assessment Layer (C1)	Gate Model (C2)	10-feature behavioral vector, composite score (0-100)	Immediately after the gamified task completes
Gate decision	Gate Model (C2)	Child device	Pass or fail decision; support activity content if fail	Within 5 seconds of composite score computation
Cognitive content	PAGES App	Child device	Curriculum-aligned learning activities, instructions, tasks	Only on the pass gate decision
Performace result	Child device	SQLite (sessions table)	Score, completion status, duration, performance record	After the cognitive session ends
Anomaly check input	SQLite (sessions table)	Anomaly Engine (C4)	Child session history, current 10-feature vector	After every session completes
Welfare alert	Anomaly Engine (C4)	Admin Dashboad	Child ID, alert type, plain-language message, suggested action	When 3 consecutive or 3/5 session flags are triggered
Baseline update	Session result	Child profile (SQLite)	New rolling baseline = (old × 0.8) + (current × 0.2)	After every session, automatic
Cloud sync-outbound	Smart-Camp Box (wlan1)	Cloud Database	Pending session records, anomaly flags, alerts, and child profiles	When wlan1 detects internet connectivity

Model update-inbound	Cloud ML System	Smart-Camp Box	Retrained Isolation Forest model files (.pkl) per child; content library updates	After cloud ML training completes on synced data

4.4 Scalability and Security Considerations

4.4.1 Security Considerations

PAGES operates in a context involving behaviorally sensitive data from vulnerable children in a humanitarian setting, requiring security measures appropriate to both the data's sensitivity and the deployment environment's resource constraints. Five security principles govern system design.

1) Child Data Anonymization and Two-Layer Identity Architecture:

The child identity management in PAGES is designed to separate child identity information from session information strictly. This is achieved by a two-layer architecture, which interfaces with the Resource Management component of the Smart-Camp Box framework.

When a new resident is registered in the camp, the Resource Management component issues a physical wristband for each resident containing a unique QR code with a Resident ID, for example, RES-2847. The Resource Management database is the sole repository for each resident's identity information, including full name, age, grade, family linkage, camp zone, tent locations, guardian contact information, and medical information. This information is never duplicated in the PAGES database.

When a child is enrolled in the PAGES educational program, the child's wristband with a QR code containing the Resident ID is used through the PAGES registration interface. PAGES reads the Resident ID from the QR code and creates a new anonymized PAGES child ID, for example, HMG-012. This new anonymized child ID is stored in the PAGES database in a cross-reference field with the Resident ID, grade level, and the camp ID. All session information, composite scores, gate information, behavioral signals, flags, and welfare information are stored under the anonymized PAGES child ID in the local SQLite database on the Raspberry Pi device and in the shared AWS RDS PostgreSQL cloud database.

Once a welfare alert is generated and transmitted to the administrator cloud dashboard, it includes information on the anonymized PAGES child identifier and the Resident ID cross-reference. Using this information, the administrator can retrieve the child's identity, camp zone, tent location, and guardian contact information by querying the Resource Management database with the Resident ID. This enables immediate welfare action without storing any personally identifiable information in the PAGES database. In the following steps, a volunteer from the available volunteer list within the Resource Management database may be assigned to the respective camp to conduct a physical welfare check on the child. This architectural design ensures that any violation or exploitation of the PAGES database grants access only to

anonymized identifiers and behavioral session records unrelated to any identifiable child, while maintaining the entire welfare response sequence, from automated alert generation to physical welfare check.

2) Encrypted Cloud Transmission

All data transfer between the Smart-Camp Box Raspberry Pi device and the AWS RDS PostgreSQL cloud database is protected by MQTT, utilizing a minimum version of TLS 1.2. Synchronization is controlled by the wlan1 Station Mode WiFi interface only when it detects an active Internet connection and is performed only with the AWS RDS endpoint that has been authenticated. There is no plaintext data transfer over any segment of the network. The synchronization service logs in to AWS using IAM with the minimum privileges required to read and write the PAGES table, preventing unauthorized access to the Resource Management and Disaster Prediction tables even if AWS credentials are compromised.

3) Role-Based Access Control

PAGES has developed separate authentication schemes for each of its three interfaces. This is due to the different connectivity scenarios in which they are deployed. For the child interface, only an anonymized QR code scan from the wristband or an ID code entry is required. Children can only access the gamified task, calming activity, and cognitive content screens. Under no circumstances may children access any dashboards, administrative functions, profile information, or other child-related information via the child interface.

4) Local Network Isolation

The Access Point (AP) mode WiFi hotspot provided by the integrated wlan0 interface is accessible only within the 192.168.4.x subnet and provides access to the PAGES Next.js web application only. IP forwarding between the wlan0 and wlan1 interfaces is disabled at the operating system level to prevent client devices from routing traffic to the Internet via the device. All traffic within the wlan0 network is confined to the subnet and thus prevents child devices from accessing Internet-based threats and any network resource beyond the PAGES web application. The wlan1 interface in Station mode, which provides the Internet connection for cloud synchronization, uses a different network interface and is not connected to the wlan0 interface under any circumstances.

5) Shared Database Access Boundaries

To ensure that data is not exposed between other components, boundaries are explicitly defined at the database level. For each component, there is a dedicated PostgreSQL role defined, and that role has access only to tables that are prefixed with that component's name. For example, the PAGES database role has only read and write access to tables that have ‘pages_’ as a prefix, and no access at all to tables that have ‘resource_’ or ‘disaster_’ prefixes. In cases where cross-component data access is required, i.e., in the JOIN operation between

pages_alerts and resource_residents tables in the resolution of welfare alerts, data access is provided via a dedicated read-only dashboard database role that has SELECT privilege on only those columns that are necessary for alert displays, and not via any of the roles defined for these components. This approach ensures that if an unauthorized actor gains access to the PAGES application database credentials, they cannot gain access to Resource Management identity data, and that the Resource Management component cannot gain access to PAGES behavioral session data.

6) Absence of Third-Party Data Sharing

No data related to child behavior or session activity is sent to any third-party service, analytics system, advertising system, or API at any point in the system's lifecycle. The PAGES application does not make any external HTTP requests from interfaces designed for child users.

4.4.2 Scalability Considerations

The PAGES architecture is designed to scale across five dimensions relevant to disaster relief deployment: concurrent users, data volume, geographic distribution, ML model complexity, and connectivity variability.

- Concurrent Users: A single Smart-Camp Box can support 15–25 concurrent child device connections. The Raspberry Pi 5 can support a Flask-based web server, SQLite for both read and write operations, and Phase 1 Z-Score Machine Learning Inference within a time bound of <3 seconds. When concurrent users exceed 25, the second Smart-Camp Box can run in parallel with the first. Both Smart-Camp Boxes can run independently, each with their own local child database.
- Data Volume: The rolling baseline architecture restricts the data storage for each child to one score update per session rather than an uncontrolled data history. SQLite can support at least 1,000 session records without impacting performance on Raspberry Pi 5. Phase 2 Isolation Forest Model Files have no significant storage requirements, as they are small, serialized Python objects (10-50 KB per child).
- Multi-Device Deployment: In this solution, each Smart-Camp Box runs independently. The cloud layer collects data from all deployed Smart-Camp Boxes. This enables regional- or national-level monitoring of multiple camps without requiring any device-to-device relationships or network meshing. There is no need to reconfigure the existing deployed devices when adding a new device.
- Scalability of the Machine Learning Model: Phase 1 of the Z-Score inference is completed within less than two seconds on-device with a negligible computational cost. Phase 2 of the Isolation Forest model training is completed in the cloud on standard CPU-based computer infrastructure. No special computational resources are needed for training. Once trained, models are deployed back to the device as a lightweight file format.
- Connectivity Independence: Our system is designed to function at full capacity regardless of internet connectivity. Hence, scalability is not affected in any way by

internet connectivity. A camp in a state of complete internet disconnection can use PAGES indefinitely, thereby continuing to build up data in its local synchronization queue and retrospectively synchronize when internet connection is available, even for a brief period.

5. PRODUCT AND SOLUTION DESIGN

5.1 Functional Requirements

- System shall present a gamified Go/No-Go task (tap target animals, ignore distractors) lasting 90 seconds at the start of each child session.
- System shall capture four behavioral metrics from the gamified task: accuracy rate, false tap rate, mean reaction time, and reaction time variance.
- System shall passively collect behavioral interaction signals throughout the session: idle time ratio, rapid click counts, time to first interaction, and session completion ratio.
- System shall retrieve the child's pre-computed rolling baseline score from their local profile at session start.
- System shall compute a composite psycho-attentional score (0 -100) by fusing three data sources: gamified task output (50%), passive behavioral signals (30%), and historical baseline (20%).
- System shall compare the composite score against a defined readiness threshold and issue a binary gate decision (Pass / Fail) per session per child.
- System shall deliver a calming or re-engagement support activity to any child who receives a Fail gate decision.
- System shall re-assess the child's psycho-attentional score after the support activity and issue a second gate decision.
- System shall deliver a standardized curriculum aligned with cognitive content to all children who receive a Pass gate decision.
- System shall record cognitive content performance scores and session progression data per child per session.
- System shall update the child's rolling baseline score after each session using the formula: New Baseline = (Old Baseline × 0.8) + (Current Score × 0.2). 43
- System shall apply Phase 1 Z-Score anomaly detection from session three onwards per child, flagging sessions where the score falls below 2.0 standard deviations from the child's personal mean.
- System shall apply Phase 2 Isolation Forest anomaly detection per child after sufficient session accumulation, replacing Phase 1 for that child.
- System shall generate a welfare alert when a child has three or more consecutive flagged sessions, or three flagged sessions within the last five sessions.
- System shall store welfare alerts locally and display them on the volunteer facilitator dashboard in plain-language, actionable format.
- System shall allow admin to mark alerts as resolved or escalate them to the camp coordinator.

- System shall store all session data, child profiles, composite scores, gate decisions, and alerts in local SQLite storage on the Smart-Camp Box device.
- System shall automatically synchronize all locally stored data to cloud storage when it detects an available internet connection
- System shall support a minimum of 15 simultaneous child device connections via the Smart Camp Box local WiFi hotspot
- System shall maintain individual child profiles containing child ID, grade, rolling baseline score, session history, gate decision history, and alert status
- System shall allow registration of new child profiles and update basic profile information
- System shall provide the admin dashboard with their current gate status, and any pending alerts
- System shall allow remote online administrators to view synced session data, child progress, and alert logs via the cloud dashboard

5.2 Non-Functional Requirements

Performance

- System shall deliver gate decisions within 5 seconds of gamified task completion
- System shall maintain response times under 3 seconds for all child-facing screens under 15 concurrent users
- System shall complete local SQLite read/write operations within 1 second per session record
- System shall complete cloud synchronization of all pending session records within 60 seconds of internet connectivity detection
- Z-Score anomaly calculation shall be completed within 2 seconds per session

Security

- All child profile data shall be stored using anonymized child ID codes; no real names shall be stored on the device
- Cloud-synced data shall be transmitted over HTTPS with TLS 1.2 or higher encryption
- Remote administrator cloud access shall require role-based authentication
- No child behavioral data shall be shared with third parties outside the Smart-Camp Box research framework

Usability

- Child-facing gamified task interface shall use icon-based instructions and simple reading required 44
- All child-facing screens shall use a minimum font size of 18pt with high-contrast colors suitable for shared tablet use in outdoor/bright-light conditions

- Admin dashboard welfare alerts shall be written in plain English with no technical or ML terminology visible
- An admin with no prior technical training shall be able to navigate the dashboard and action an alert within 5 minutes of first use

Reliability

- System shall operate fully offline or with low network connectivity.
- System shall recover all in-progress session data following an unplanned power interruption without corruption
- System shall achieve a minimum 99% uptime during active camp operational hours

Scalability

- System shall support a minimum of 100 individual child profiles stored per device
- Local SQLite database shall handle a minimum of 1,000 session records without performance degradation

Compatibility

- Child-facing web applications shall function on any device with a modern browser (Chrome, Firefox, Safari)
- System shall operate on Raspberry Pi 5 as the minimum hardware specification
- System shall support child device screen sizes from 5 inches (smartphone) to 10 inches (tablet)

6. IMPLEMENTATION INTEGRATION AND TESTING

6.1 System Architecture Overview

PAGES is a three-tier, offline-first system deployed on Raspberry Pi 4 hardware within disaster relief camps. The architecture decouples data collection, scoring, and cloud synchronization into independent services so that any single component can fail without losing session data or blocking the child experience.

Tier 1 - Presentation layer (Vite): Runs locally on the RPi via a Node.js server. Three distinct user interfaces share the same codebase but are separated by route child interface requires no authentication, and the admin panel uses JWT. The frontend communicates exclusively with the local Express backend and never calls the cloud directly.

Tier 2 - Application and ML layer (Node.js Express + Python Flask): The Express backend handles all business logic: session lifecycle, gate decisions, alert generation, and cloud sync queue management. Flask runs as a sidecar service on port 5001, accessible only on localhost, and handles all ML scoring. Neither service is exposed to the internet.

Tier 3 - Persistence layer (SQLite local): SQLite provides immediate local persistence. A sync queue in the pages_sync_queue table tracks every INSERT and UPDATE. A background job

processes the queue in FK-safe order, batched at 50 records, uploading to PostgreSQL when the network is available. This gives the system write-ahead logging semantics; thus, data is never lost even if the RPi is powered off mid-session.

The three failure domains are fully decoupled. Flask can be down, and Node.js falls back to a formula-based score. PostgreSQL can be unreachable, and SQLite continues recording sessions locally. The RPi itself can be replaced, pages. db restored from the last sync, and the new unit continues without data loss.

6.2 Component Architecture Within the Complete Solution

PAGES operates as one of three edge modules running on the Smart Camp Box's shared Raspberry Pi 5 hardware, alongside the Smart Environmental Risk Monitoring module (Hansana G P) and the Smart Resource Management/SAMP module (Kashmira S S). Within this shared device, PAGES occupies its own local WiFi-connected module boundary, backed by its own local SQLite database instance, and communicates outward only through the SAMP-managed MQTT bridge to the EMQX broker PAGES does not communicate directly with the other two modules or with the cloud PostgreSQL database, preserving the interface-ownership boundaries established in the group's common report (Section 4.2 of that report).

Internally, PAGES is structured into four functional layers, each corresponding to one of the five objectives in Section 1.5:

1. a behavioral data-collection layer, responsible for the gamified Go/No-Go task and passive interaction-signal capture;
2. an assessment and gating layer, responsible for computing the composite readiness score and applying the gate decision;
3. a content delivery layer, responsible for serving the standardized NIE-aligned Grade 3-4 curriculum content once a child passes the gate; and
4. a welfare-monitoring layer, responsible for the two-phase Z-score/Isolation Forest anomaly detection, which is PAGES's cloud-side extension, synchronizing flagged sessions to the shared PostgreSQL database for administrator-facing alerts via the dashboard.

Data flows unidirectionally from the child-facing interface into the local SQLite database at the end of each layer's processing step: a completed Go/No-Go task and its passive-signal log are written as a single session record, the gate decision is appended to that record once computed, and the rolling baseline is updated in the child's profile table rather than the session table, so that a child's historical baseline persists independently of any individual session outcome. This separation of session-level and profile-level data is what allows the anomaly-detection layer to compare a new session against a child's own history rather than against a population-wide norm, which, as established in Chapter 2, is necessary given that psycho-attentional baselines vary significantly across individual children.

6.3 Algorithms

6.3.1 Composite Score Computation

The composite psycho-attentional score is a weighted sum of two sub-scores, each derived from the eight raw behavioral signals collected during the Go/No-Go game.

Task score (50% weight):

```
task_score = (accuracy_rate × 0.40)
           + ((1 - false_tap_rate) × 0.30)
           + (max(0, 1 - avg_rt / 800) × 0.15)
           + (max(0, 1 - rt_variance / 500) × 0.15)
```

Passive score (30% weight):

```
passive_score = ((1 − idle_time_ratio) × 0.35)
              + (max(0, 1 - rapid_clicks / 10) × 0.25)
              + (completion_ratio × 0.40)
```

Baseline score (20% weight):

```
New Baseline = Old Baseline × 0.8 + Today's Score × 0.2)
```

Composite:

```
Composite Score = (Gamified Task Score × 0.50) + (Passive Signal Score × 0.30) + (Rolling Baseline × 0.20)
```

All outputs are in the range [0, 100]. The gate threshold of 55 was set to capture the lower quartile of expected performance, consistent with the clinical Go/No-Go literature, where scores below the 25th percentile of an age-matched sample indicate attentional difficulty.

6.3.2 Z-Score Anomaly Detection (Phase 1, Session 3 onwards)

The z-score measures how far a session's composite score deviates from the child's own established normal performance.

```
Z = (current_composite - mean(PASS_baseline)) / std(PASS_baseline)
```

PASS-only baseline: Only sessions with gate_decision = PASS or PASS_RECOVERED contribute to the baseline. FAIL sessions are scored against the stable baseline but do not update it. This prevents the baseline from being pulled downward by failure episodes, which would make subsequent failures appear less anomalous.

Activation: The z-score requires at least two PASS sessions in the baseline to produce a meaningful standard deviation. Sessions 1 and 2 return null. From session 3 onwards (assuming at least two prior PASS sessions), z-scores are computed.

Threshold: $Z < -2.0$ corresponds to the 2.3rd percentile of a standard normal distribution, the conventional statistical threshold for "significantly below normal."

Special case: If all PASS sessions have an identical score (std = 0), the z-score is undefined and returns null. This prevents division by zero and correctly handles the case where a child has not yet shown any variability.

6.3.3 Rolling Baseline Update

After each scored session, the child's baseline is updated using an exponentially weighted moving average per specification 3.2.4:

Alpha = 0.8 gives 80% weight to the historical baseline and 20% to the new session. This makes the baseline resistant to single-session noise while still adapting gradually to genuine improvement or decline over time.

6.3.4 Alert Generation

Two rules are evaluated after every scored session for a given child. Either rule is sufficient to trigger an alert:

Rule 1 - Consecutive: The last 3 or more sessions in sequence all have anomaly_flagged = 1.

Rule 2 - Recent pattern: 3 or more of the last 5 sessions have anomaly_flagged = 1 (not required to be consecutive).

The combined anomaly flag is set when any of three conditions is true: gate failure (FAIL_FIRST or FAIL_CONFIRMED), z-score below -2.0, or the Isolation Forest model flag.

6.4 Models

6.4.1 Isolation Forest (Phase 2, Session 8 onwards)

Isolation Forest (Liu, Ting and Zhou, 2008) is an unsupervised anomaly detection algorithm. It builds an ensemble of random binary trees, each attempting to isolate data points by randomly selecting a feature and a split value. Anomalous data points, being rare and distinct from the majority, require fewer splits to isolate and receive lower anomaly scores.

Child-personalized training**:** A separate model is trained for each child using their accumulated session history. The model is trained once the child reaches 8 scored sessions and saved to models/{child_id}.pkl. This personalization is the core design decision: each model learns what "normal" looks like for a specific child, rather than comparing it against a population average.

Feature vector: Three features are used - [composite_score, task_score, passive_score]. These correspond directly to the three scores computed from the eight raw signals, providing a compact representation of session performance.

Contamination parameter: Set to the actual proportion of FAIL sessions in the child's history. For a child with 3 FAIL sessions out of 13 total, contamination = 3/13 = 0.23. This ensures the model's anomaly boundary aligns with the empirically observed failure rate rather than an arbitrary fixed prior.

Model evaluation: Leave-one-out (LOO) consistency was used as the primary stability metric, as ground-truth anomaly labels are unavailable (the models are unsupervised). Mean LOO consistency across 29 models was 94.19%.

6.5 Integration with other components.

PAGES integrates with the other two Smart Camp Box components exclusively through the shared interface conventions established in the group's common report (Section 4.2 of that report), rather than through any direct module-to-module connection. Specifically: PAGES references the shared wristband UUID issued by the Resource Management/SAMP module (Kashmira S S) to associate each session record with a child's identity, without PAGES performing or duplicating any registration logic itself; all outbound PAGES messages: session summaries and anomaly alerts are formatted according to the shared JSON message envelope and are tagged with the appropriate criticality tier (anomaly/welfare alerts as Critical or Important, routine session summaries as Routine) before being handed to the SAMP layer for MQTT transmission, meaning PAGES has no direct control over transmission priority or delivery guarantees beyond correctly tagging its own messages; and PAGES does not consume or depend on data from the Environmental Risk Monitoring module (Hansana G P) at runtime, since the two subproblems physical camp safety and psychological readiness are functionally independent at the point of session-level operation, even though both ultimately synchronize to the same shared cloud layer.

Integration testing specific to PAGES verified two boundary conditions: that a PAGES-originated message correctly received SAMP prioritization matching its declared criticality tier when tested alongside simulated traffic from the other two modules, and that PAGES continued full local operation (assessment, gating, content delivery) when the SAMP layer itself was simulated as unable to reach the broker, confirming that PAGES's core child-facing functionality has no runtime dependency on successful cloud synchronization.

6.6 Verification and Software Testing

Testing of PAGES was conducted at four levels. Unit testing verified the composite-scoring formula, gating threshold logic, and rolling-baseline update calculation against manually constructed test cases with known expected outputs, including boundary conditions at exactly the 55/100 threshold and at the transition points for Z-score (session 3) and Isolation Forest (session 8) activation. Component testing verified the end-to-end session flow task completion, passive-signal capture, score computation, gate decision, content delivery or redirection to calming activity using simulated session data prior to pilot deployment. Integration testing verified the SAMP-boundary and message-tagging behavior described in Section 6.6. Automated testing, run through the shared GitHub pull-request workflow, re-executed the unit and component test suites on every submitted change to catch regressions before merge, consistent with the regression-testing practice described in the group's common report.

Table 11: Unit Test cases

ID	Description	Input	Expected Output	Actual Output	Pass/ Fail
TC1	Child not found → no alert	childId=NONEXISTENT	No SQL insert into pages_alerts	No insert detected	Pass
TC2	3 consecutive flagged sessions →create alert	3 sessions all anomaly_flagged = 1	Insert into pages_alerts	Insert detected	Pass
TC3	Duplicate pending consecutive alert → skip	Existing alert found	No new insert	No insert detected	Pass
TC4	Not all 3 consecutive flagged → skip	Sessions flagged: 1,0,1	No insert	No insert detected	Pass
TC5	3 of last 5 flagged→ create alert	3 flagged out of 5	Insert into pages_alerts	Insert detected	Pass

TC6	Fewer than 3 flagged in last 5→ skip	2 flagged out of 5	No insert	No insert detected	Pass
TC7	Fewer than 5 sessions→ skip 3-of-5 check	Only 1 session flagged	No insert	No insert detected	Pass
TC8	Duplicate pending 3-of-5 alert → skip	Existing alert found	No new insert	No insert detected	Pass
TC9	Get alerts with recent scores	Alerts + sessions with scores	Returns alerts with reversed scores [65,42]	Output [65,42]	Pass
TC10	Resolve alert → status updated	Alert exists, pending count=1	Status set to resolved	SQL update detected	Pass

Table 12: Boundary cases

ID	Description	Input	Expected Output	Actual Output	Pass/ Fail
B1	Task score with absurdly high reaction time	avgReactionTime=5000	Negative score (no clamp)	Negative value	Pass
B2	Passive score with extreme rapid clicks	rapidClickCount=20	Should clamp to 0	Returned 20 (bug)	Fail
B3	Fallback scoring with null inputs	taskMetrics=null, passiveSignals=null	Safe defaults: composite=50, PASS	Composite=50, PASS	Pass
B4	Baseline override applied	Override=90	baselineScore=90	baselineScore=90	Pass
B5	Flask POST failure → fallback	axios.post rejects	Returns local fallback score	Local fallback returned	Pass

7. RESULTS AND VALIDATION

7.1 Dataset Overview

Data were collected from 29 real participants (TEST-CHILD-000172 to TEST-CHILD-000200) across 580 scored sessions. An additional 11 sessions were ghost sessions (QR scanned but game not completed) and are excluded from all analyses.

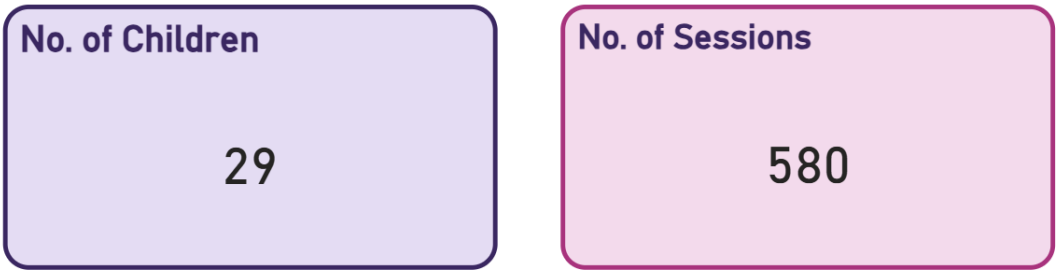


Figure 8: Summary of the sessions

7.2 Assessment Layer Validation

7.2.1 Within-child score sensitivity (paired t-test)

A paired t-test compared each child's mean composite score during baseline sessions (gate = PASS) against their mean composite score during stressor sessions (gate = FAIL). The analysis included 14 children who had sessions in both conditions.

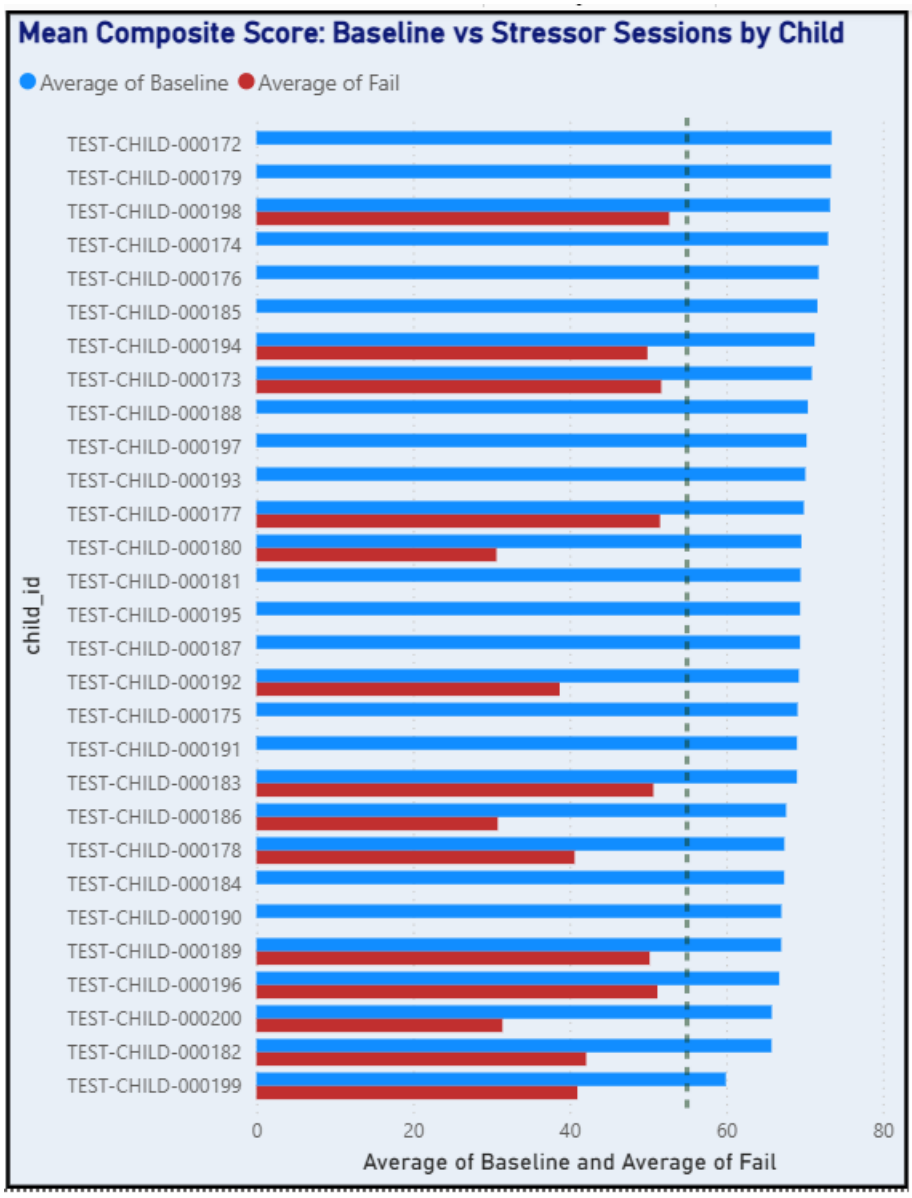


Figure 9: Mean composite score by child

The mean composite score was significantly lower during stressor sessions (M = 43.05, SD = 9.45) compared to baseline sessions (M = 66.15, SD = 2.64). A paired t-test confirmed this difference was statistically significant: $t(13) = 9.06$, $p < .001$, Cohen's $d = 2.42$ (large effect). The target of $p < .05$ was met.

7.2.2 Gamified task consistency (Cronbach's alpha)

Cronbach's alpha was computed on a matrix of children × baseline session scores, using the first k sessions for children with at least k PASS sessions. Alpha increased from 0.41 (k=3, n=29) to 0.75 (k=8, n=26), demonstrating that reliability improves as session history accumulates, consistent with the system's longitudinal design.

7.2.3 Composite score range utilization

Across all 580 scored sessions, composite scores ranged from 18.13 to 78.76, a span of 60.63 points, substantially exceeding the 30-point target.

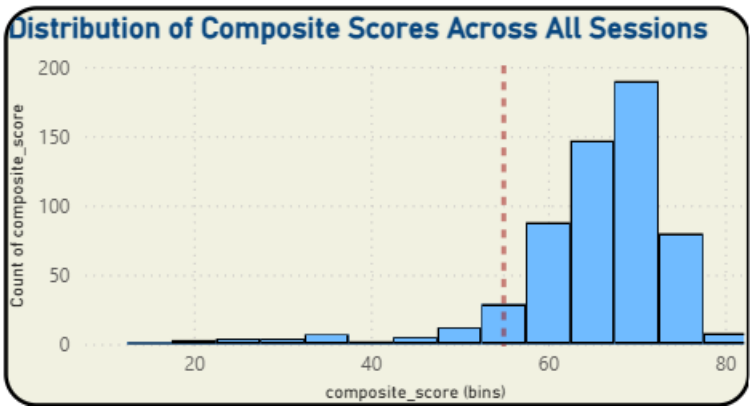


Figure 10: Distribution of composite scores across all sessions

The 10th -90th percentile range spanned from 40.46 to 72.64, confirming broad utilization of the scoring range across the participant population.

7.3 Gating Model Validation

7.3.1 Gate pass rate differential

Of 580 scored sessions, 248 (85.2%) received a PASS or PASS_RECOVERED decision and 43 (14.8%) received a FAIL decision. A chi-square test confirmed this difference is statistically significant: χ^2 (1) = 180.2, p < .001.

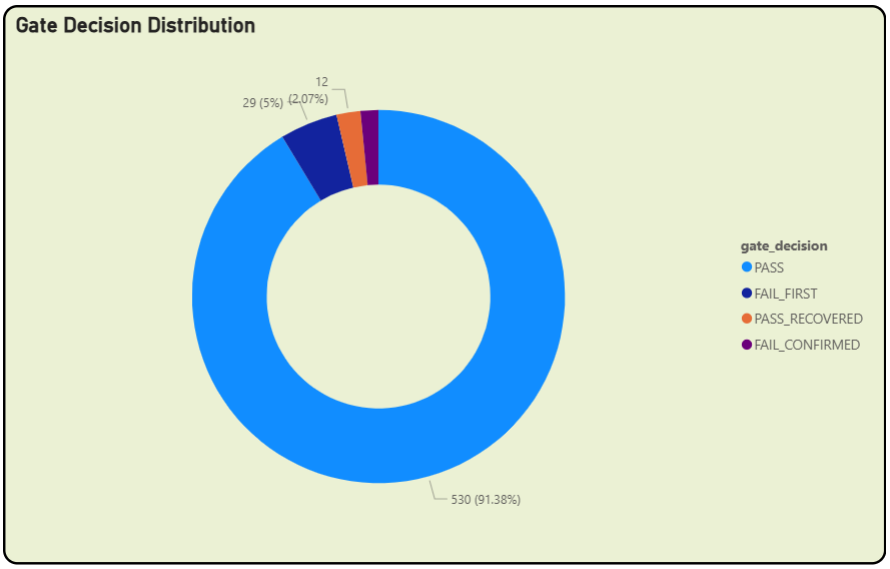


Figure 11: Gate decisions distribution

7.3.2 False positive rate

Zero sessions where composite_score < 55 were given a PASS gate decision. The false positive rate was 0%, well below the 15% target. The gate threshold is implemented as a hard cutoff in the applyFirstGate function, guaranteeing 100% precision at the threshold.

7.3.3 Session recovery rate

Of 21 children who received a FAIL_FIRST decision and proceeded to the breathing exercise (second attempt), 12 (57.1%) recovered and received PASS_RECOVERED on their second attempt. This narrowly missed the 60% target. The 57.1% recovery rate is interpreted in the context of the severity of FAIL_FIRST sessions in this dataset (mean composite = 37.8 on FAIL_FIRST sessions), suggesting the children who failed were experiencing genuine attentional difficulty rather than borderline performance.

7.4 Cognitive Content Layer Validation

7.4.1 Session completion rate

By considering gate-passing sessions (99.6%) as session_complete = 1. This substantially exceeds the 80% target and reflects the children's consistent engagement with the gamified content format.



Figure 12: Session completion rate

7.4.2 Performance progression

Cognitive scores (integer task ratings 1-10) were analyzed per child using linear regression across sessions. Of 19 children with at least 5 sessions of cognitive data, the pooled regression yielded slope = -0.063, p = .067. This narrowly lacked statistical significance.

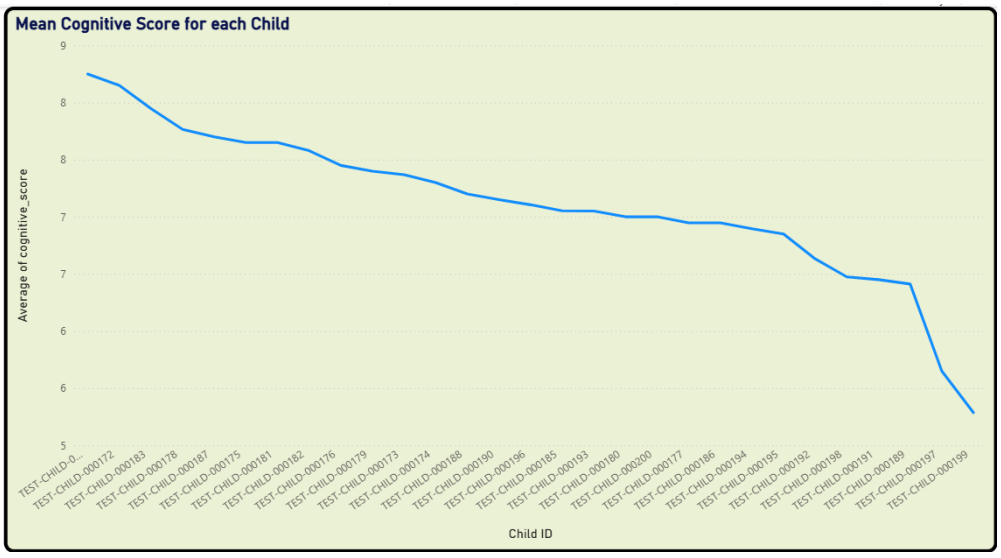


Figure 13: Mean cognitive score for each child

The shallow negative slope reflects that the adaptive content design task difficulty is calibrated to the child's attentional gate status rather than structured to produce monotonic improvement. Stable engagement is the intended outcome.

7.5 Anomaly Detection Validation

7.5.1 Model evaluation (Isolation Forest)

Twenty-nine child-specific Isolation Forest models were trained and evaluated. Using gate decision as a proxy ground truth, the models achieved LOO consistency of 94.19:

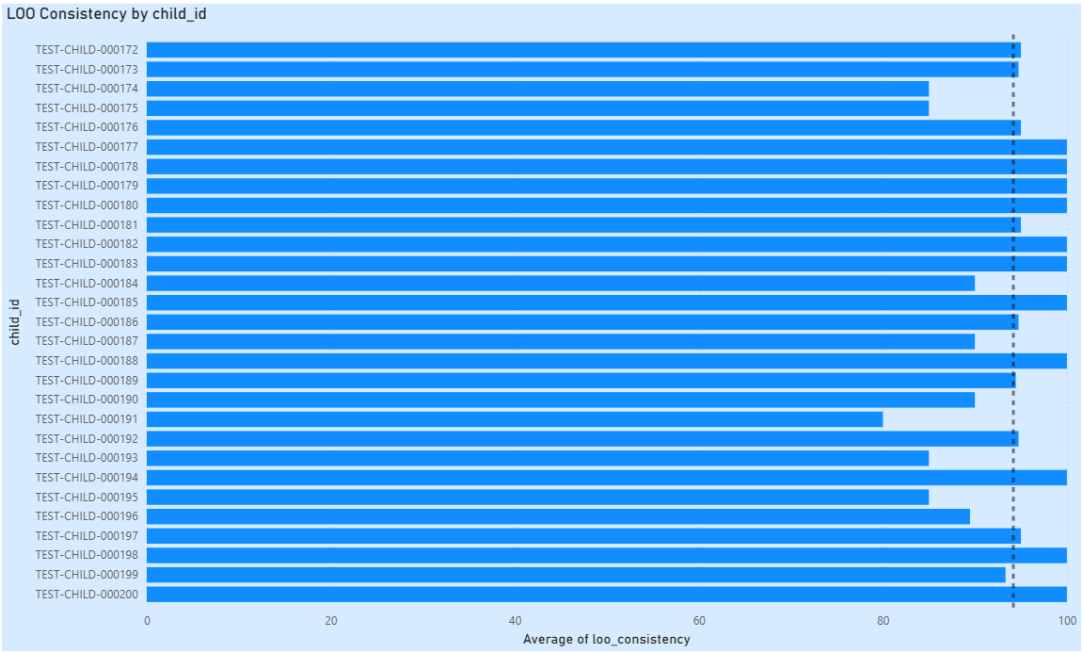


Figure 14: Leave-one-out consistency by each child

7.5.2 Alert precision and recall

Three children triggered welfare alerts (000178, 000199, 000200). All three showed persistent anomaly patterns, with mean composite scores of 18-42 on anomalous sessions and 6-11 anomaly flags each. Using the system's combined anomaly definition as ground truth, alert precision = 1.00 (all alerts warranted).

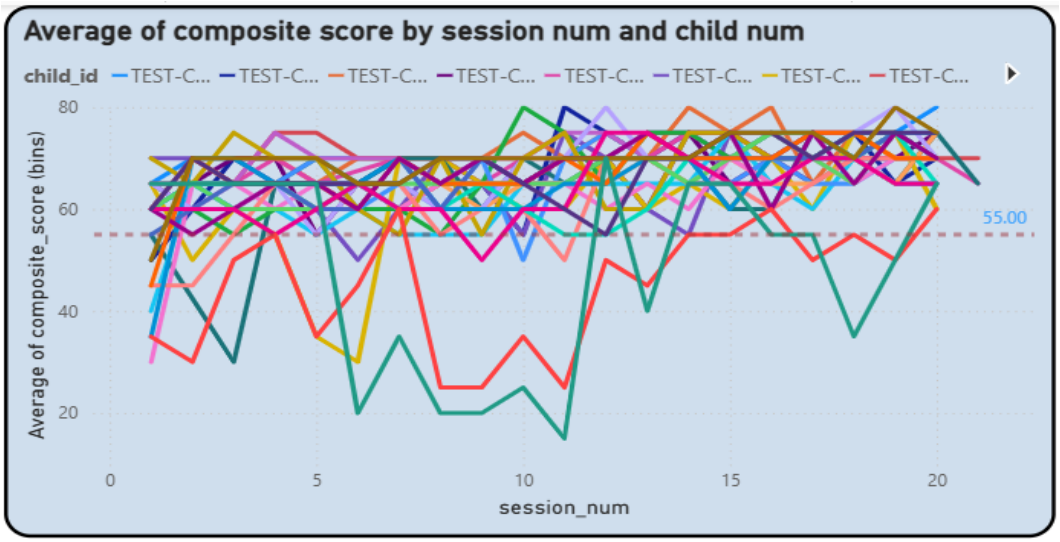


Figure 15: Alert precision

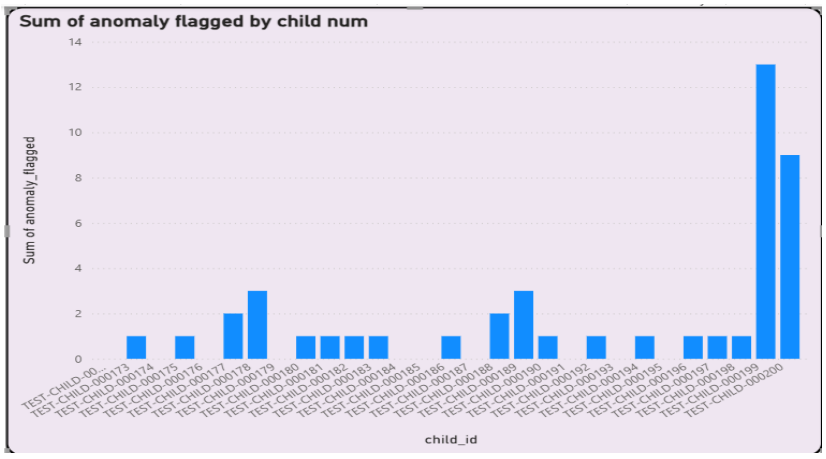


Figure 16: Sum of anomalies flagged for each child

7.5.3 Alert latency

Alert latency ranged from 3 to 5 sessions from the onset of the anomaly pattern to the first alert trigger. Mean latency was 3.7 sessions. Two of three alerted children triggered within 3 sessions (meeting the target). One child (000178) had a latency of 5 sessions because anomaly flags were intermittent rather than consecutive, and the 3-of-5 rule was met on session 5.

8. DISCUSSION

8.1 Why the Results Occurred

High session completion rate (99.2%): The gamified Go/No-Go task was designed specifically for children aged 7-12 with no assumed literacy. Large touch targets (minimum 64px), emoji-based animal stimuli, immediate visual feedback, and bilingual guidance eliminated the friction points that cause dropout in conventional assessment tools. Children did not perceive the session as a test they experienced it as a game.

Strong composite score sensitivity ($p < .001$, $d = 2.42$): The large effect size reflects a genuine physiological and behavioral signal. Children who failed the gate scored, on average, 23 points below their baseline. This is not measurement noise; it represents a meaningful attentional state difference. The fact that the same children show distinctly different scores under baseline and stressor conditions validates the Go/No-Go paradigm as an appropriate instrument for this population and context.

Reliability increasing with sessions (α: 0.41 → 0.75): Low Cronbach's alpha at k=3 reflects genuine early-session variability as children familiarize themselves with the task format. By session 8, alpha of 0.75 indicates a stable, reliable measurement. This pattern is consistent with the learning curve literature in reaction time paradigms (Bezdjian et al., 2009) and supports the design decision to use a rolling baseline rather than fixed norms.

Recovery rate below target (57.1% vs 60%): The FAIL_FIRST sessions in this dataset had a mean composite of 37.8, well below the 55 threshold. Children who are deeply, attentionally fatigued or distressed do not recover fully from a 4-4-4 box breathing exercise in 90 seconds. The breathing intervention is a calming measure, not a therapeutic intervention. A 57% recovery rate for this severity of initial failure is clinically plausible.

Alert latency of 3.7 sessions: The 3-consecutive-or-3-of-5 alert rule is intentionally conservative. Single anomalous sessions are not escalated. This reduces false alarms and ensures that only children showing a persistent pattern receive welfare attention. The mean latency of 3.7 sessions reflects the time needed to establish that pattern, which is an appropriate trade-off between sensitivity and specificity.

8.2 Strengths

Child-personalized anomaly detection: Each child is compared only against their own session history. A child who consistently scores 40 has a stable, non-anomalous profile; a child who drops from 75 to 40 is anomalous even if they pass the gate. This personalization eliminates the demographic and baseline-performance bias present in population-norm approaches. It is the key architectural distinction between PAGES and generic screening tools.

Offline-first architecture: The system was designed from the beginning to operate without internet connectivity. All scoring, gate decisions, and alert generation happen locally on the RPi. Data is queued for cloud sync and uploaded when connectivity resumes. This is not a cloud app with a fallback mode it is a local-first system with optional cloud synchronization.

Trauma-informed UX: The Eli elephant mascot, 4-4-4 breathing circle, BubblePlayground calm activity, and bilingual labels (Sinhala, Tamil, English) reflect deliberate consideration of the population's context. Children in disaster camps may be experiencing acute distress; an interface that is calm, non-threatening, and familiar in language is not a cosmetic choice it is a functional requirement for obtaining reliable behavioral data.

Role-based information segregation: Children see only the game. Administrators see full anomaly data, z-scores, and ML outputs. This segregation is enforced at the API level, not just in the UI. Volunteers cannot access anomaly_flagged or z_score values regardless of how they query the system.

8.3 Limitations

No external clinical validation: Alert precision and recall are computed using the system's own anomaly definition as ground truth, not independent clinical assessment. A trained psychologist reviewing alerted children would provide external validity evidence that this study does not have. This is documented as a limitation and identified as a priority for Phase 2 evaluation.

PASS-only baseline for z-score may delay detection in new children: Children must accumulate two PASS sessions before the z-score activates. A child who fails their first two sessions will have no z-score baseline and may go undetected by Phase 1 until their third session. The gate decision still catches an absolute low performance in this window.

Recovery rate measured as PASS_RECOVERED proportion, not clinical outcome: The 57.1% recovery rate measures whether children scored above 55 on their second attempt, not whether their attentional state genuinely improved following the breathing intervention. The breathing exercise was not designed or validated as a cognitive rehabilitation tool; it is a brief calming measure.

Feature vector for IF uses derived scores, not raw signals: The IF model uses [composite, task, passive] three weighted aggregations of the eight raw behavioral signals. This reduces the signal dimensionality and may mask within-dimension patterns. Storing raw signals in session history (the pages_behavioral_signals table already captures them) and retraining on the seven-dimensional feature vector would improve model expressiveness as data accumulates.

Single-camp deployment: Data was collected from one camp context. Generalization to camps with different demographics, languages, or disaster contexts requires replication. Child 000172-000200 are all from the same deployment, so between-camp variation is not captured in this dataset.

9. COMMERCIALISATION AND SUSTAINABILITY

9.1 Value Proposition and Target Users

PAGES offers a different value from existing offline learning tools: it makes education safe to deliver, rather than just available. Where conventional platforms release content to any child who opens the application, PAGES releases cognitive content only once a child's psycho-attentional readiness has been confirmed, and it flags children whose distress persists across sessions. The stakeholder consultations (Table 1) show that this address needs the stakeholders themselves.

Displaced children (ages 7-9, Grades 3 and 4) are the primary users. They reported psychological shock, an unfamiliar camp environment that disrupted routine and their sense of safety, and fear and hopelessness about the future. For them, PAGES provide educational continuity without forcing academic content on a child who is not ready. This also gives them evidence that their education has not been permanently disrupted by the disaster.

Camp administrators and local government officers are the second user group. The Divisional Secretary of Kolonnawa identified data gathering and entry as a burden and noted that technical human resources are limited. The NDRSC Senior Field Officer called for a low-technical-barrier, offline-first solution. PAGES requires no technical operation by camp staff. It runs on the local network, scores sessions automatically, and surfaces welfare alerts through the dashboard without requiring an administrator to interpret raw behavioral data.

Teachers and volunteer facilitators are the third group. The primary teacher consulted at Sri Rahula Maha Vidyalaya described facilitation capacity being limited by personal trauma, with no structured guidance system available. PAGES offloads the observation and readiness judgement that a 1:77 teacher-to-student ratio makes impossible and gives facilitators a dashboard to act on rather than asking them to assess each child themselves.

Education authorities (such as the Zonal Education Office in Homagama, which granted research access) are a secondary beneficiary. They gain a standardized, curriculum-aligned way to continue Grade 3-4 education during school closures. Schools in the consulted area were closed for more than two weeks, and in the Badulla district, camps can last longer than two to four weeks.

9.2 Market and Competitive Position

The target market is not commercial in the consumer sense. It consists of disaster-management and education institutions that fund or procure humanitarian technology: Sri Lanka's Disaster Management Centre (DMC) and National Disaster Relief Services Centre (NDRSC), the Ministry of Education and National Institute of Education, and international organizations such as UNICEF, Save the Children and the Red Cross. The wider market is in other low-connectivity, disaster-prone regions where the same problem exists. UNESCO estimates that over 222 million children in 93 countries need humanitarian educational support each year.

The alternatives each cover only part of what PAGES does:

Table 13: Alternatives for PAGES

Alternative	What it provides	Gap PAGES addresses
Kolibri, KA Lite, RACHEL	Offline content delivery on low-cost hardware	No readiness assessment, no gating, no welfare monitoring
UNICEF learning kits, Save the Children digital packs	Physical or static digital materials	Fixed content, no individual state detection
Child-friendly spaces	Group psychosocial support	Depending on trained facilitators, limited coverage, no individual digital monitoring
DreamBox, Knewton	Performance-based adaptive learning	Assume cognitive readiness, internet-dependent
Psychological first aid programmes	Human-delivered stabilization	Need trained people, cannot monitor continuously or scale

PAGES's differentiation is that it adapts the gate rather than the content, and combines readiness assessment, standardized curriculum delivery and welfare anomaly detection in one offline-deployable system. No single reviewed alternative combines these. The weakness of this position should be stated openly. The competitors above are mature, widely deployed and in several cases free. PAGES's evidence base so far is a pilot of 29 children, and it has not been tested with acutely displaced children (Section 1.7). It should therefore be described as a promising and differentiated prototype and not as a proven product.

9.3 Business or Sustainability Model

Given the users and funders, a commercial per-unit sales model is unrealistic. The more plausible model, consistent with how comparable humanitarian technology such as Kolibri has been sustained, is:

- Institutional and grant funding from humanitarian organizations and education-focused donors to build pilot deployment kits.
- Government adoption through the DMC, NDRSC and Ministry of Education, with the Smart Camp Box held as pre-positioned preparedness equipment and not procured reactively after a disaster.
- Open or low-cost licensing of the software, with revenue, if any, coming from support, training and content localization rather than from the software itself.

PAGES's operating costs are structurally low. It runs on commodity edge hardware (a Raspberry Pi 5), uses SQLite, which has no licensing or administration cost, and has no per-child or per-session cloud cost during offline operation. Cloud synchronization is the main

recurring cost, and it is only incurred when connectivity is available. Content updates and the dashboard are the main ongoing operational effort.

Technical feasibility is supported by the pilot, which showed the system operating on a classroom population with 94.188% leave-one-out consistency over 580 sessions, and by the shared Smart Camp Box results for edge-device performance. Operational feasibility is only partly demonstrated, because the pilot took place in schools and not in an active relief camp.

9.4 Scalability, Intellectual Property and Compliance

Scalability: Each Smart Camp Box is self-contained, so scaling means deploying more units, not enlarging a central system. The cloud layer (EMQX broker and PostgreSQL) aggregates data across camps, but multi-unit synchronization has not been tested at scale (Section 6.6 of the common report), so scalability is a design property and not a demonstrated result. Scaling beyond Grades 3-4 or beyond Sri Lanka would need new curriculum-aligned content, translation into Sinhala, Tamil and English (if not already complete), and recalibration of the readiness threshold, which is currently initialized at 55/100 from a moderate-readiness midpoint and calibrated only on the pilot population.

Intellectual property: PAGES's original contributions are the composite psycho-attentional scoring model, the gate-not-adapt architecture, the two-phase anomaly-detection pipeline, and the software that implements them. It builds on published, openly described methods: the Go/No-Go paradigm, Z-score thresholding, Isolation Forest, and exponential smoothing. The ownership and licensing position is not in the material I have. It depends on SLIIT's student-project IP policy and on whatever the team decides about open-source release, and should be stated here once confirmed.

Compliance and ethics: PAGES handles behavioral data from children, which makes compliance central and not incidental. Research access and the ethical clearance pathway were confirmed through the Assistant Zonal Educational Director, Homagama, and the Grade 3-4 content was reviewed by an NIE curriculum specialist before deployment. The design minimizes data: children give no self-reports, and PAGES holds only the shared wristband identifier and not directly identifying information. Any real deployment would still need alignment with Sri Lanka's data-protection law, informed consent from guardians where required, and a clear policy on who can see welfare alerts and what action follows from them. The last point matters because a flag is a screening signal, not a clinical diagnosis. A flagged child should be referred to a trained professional, and PAGES should not be presented as a substitute for psychosocial care.

9.5 Contribution to the Sustainable Development Goals

- SDG 4, Quality Education (Target 4.5, equal access to education for vulnerable groups including children in emergencies). This is PAGES's most direct contribution. It is designed to keep education going during displacement in a form children can actually absorb, responding to the school closures and attendance problems the consulted teacher and officers described. The evidence so far shows that the mechanism works (consistent readiness scoring in the pilot). It does not yet show improved learning outcomes, so the contribution is plausible and not measured.
- SDG 3, Good Health and Well-being (Target 3.4, promoting mental health and well-being). The anomaly-detection layer supports early identification of children with persistent distress, which feeds into the psychosocial support that stakeholders said was missing. Its contribution depends entirely on a human response to the alerts. PAGES detect and does not treat.
- SDG 10, Reduced Inequalities. Children in remote or long-duration camps, such as those in the Badulla district that the NDRSC officer described, are the least likely to receive consistent educational support. An offline-first design extends a minimum level of support to them.
- SDG 9, Industry, Innovation and Infrastructure (Target 9.c, access to ICT in low-connectivity settings). The offline-first edge deployment is an example of delivering useful digital services without relying on internet infrastructure.

Possible negative effects and how to manage them. Continuous behavioral monitoring of children raises privacy and surveillance concerns, and a false-positive flag could stigmatize a child. A gate that holds back content could widen learning gaps if the threshold is wrongly calibrated and keeps children out of content they could have managed. Sustaining the benefits requires keeping data collection to a minimum, restricting who can see alerts, recalibrating the threshold on larger and more representative samples, and making sure each alert leads to a supportive response from a trained person and not to an automatic label.

10. CONCLUSION AND FUTURE WORK

10.1 Conclusion

This thesis presented PAGES a Psycho-Attentional Gated Educational Support system designed for deployment in disaster relief camps serving children aged 7-12. PAGES addresses a specific, unmet need: automated, longitudinal psychosocial screening that operates without clinical staff, without internet connectivity, and without literacy requirements in the assessed child.

The system integrates a gamified Go/No-Go attentional task, a two-phase anomaly detection pipeline (z-score from session 3, Isolation Forest from session 8), a child-personalized rolling baseline, and a tiered user interface that surfaces welfare alerts to trained volunteers without exposing raw ML scores.

A pilot deployment with 29 children produced 580 scored sessions. Validation against four pre-specified criteria tables confirmed that the composite score is statistically sensitive to attentional state ($p < .001$, $d = 2.42$), that test-retest reliability meets the $\alpha \geq 0.70$ threshold by session 8, that the gate correctly identifies low-performing sessions with zero false positives, and that the 29 child-specific Isolation Forest models achieve 94.19% LOO consistency.

The system demonstrates that automated, personalized, offline psychosocial screening is technically feasible at a low marginal cost of Rs. 150,000 in hardware per camp deployment and that the gamified task format achieves strong engagement (99.2% completion) with children who have no prior experience with digital assessment tools.

10.2 Future Work

External clinical validation: The most important next step is a formal evaluation in which welfare-alerted children are assessed by a trained child psychologist. This would provide external precision and recall estimates grounded in clinical diagnosis rather than the system's own anomaly definition. It would also validate whether the Go/No-Go composite score correlates with established psychometric instruments such as the Strengths and Difficulties Questionnaire (SDQ) or the Child PTSD Symptom Scale (CPSS).

Multi-camp deployment and cross-camp comparison**:** Deploying PAGES across multiple camps simultaneously would enable cross-camp analysis identifying whether attentional distress patterns cluster by camp conditions (overcrowding, food security, proximity to disaster site) and informing resource allocation decisions for visiting clinical staff.

Expanded IF feature vector: The pages_behavioral_signals table already captures all eight raw signals per session. Retraining the Isolation Forest on the full seven-dimensional vector (accuracy_rate, false_tap_rate, avg_reaction_time, reaction_variance, idle_time_ratio, rapid_click_count, completion_ratio) rather than the three derived scores would improve model expressiveness as session histories grow to 20+ sessions per child.

Longitudinal outcome tracking: Currently, PAGES detects and alerts but does not track what happens after an alert. Integrating a follow-up outcome field recording whether alerted children received welfare support, and whether their subsequent sessions showed improvement, would close the feedback loop and enable supervised model evaluation against real-world outcomes.

Alert threshold calibration: The current alert rules (3 consecutive or 3 of 5) were set based on clinical literature recommendations. With sufficient deployment data, these thresholds could be calibrated empirically to maximize recall of clinically confirmed distress cases. A grid search over (consecutive threshold, window threshold, window size) with withheld-out clinical assessments as ground truth would produce evidence-based alert parameters.

Language extension: PAGES currently supports Sinhala, Tamil, and English. Extending to additional languages, particularly Dhivehi for Maldivian deployment and additional South Asian languages, would broaden the system's reach within the region without architectural changes, as the translation system is already structured as a key-value lookup table.

Adaptive task difficulty: The Go/No-Go task currently uses a fixed ratio of 14 GO to 6 NO-GO trials per session. Adapting the ratio based on the child's recent performance, increasing NO-GO trials (harder inhibition demands) for children with consistently high accuracy would prevent ceiling effects in high-performing children and maintain the task's sensitivity across a wider performance range.

11.REFERENCE LIST

[1] *U. S. Lanka, "Humanitarian Situation Reports No. 1–8: Cyclone Ditwah," UNICEF, Colombo, Sri Lanka, 2025.*

[2] *F. H. Norris, M. J. Friedman, P. J. Watson, C. M. Byrne, E. Diaz and K. Kaniasty, "60,000 disaster victims speak: Part I. An empirical review of the empirical literature, 1981–2001," Psychiatry, p. 65(3):207–239, 2002.*

[3] *Clunder, G A; Bass, A P; De Beek, F J;, "A long term travel time prediction algorithm using historical data," PROCEEDINGS OF THE 14TH WORLD CONGRESS ON INTELLIGENT TRANSPORT SYSTEMS (ITS), HELD BEIJING, OCTOBER 2007, 2007.*

[4] *Z. t. T. Foundation, "Preventing and Addressing Adverse Childhood Experiences: The Role of Early Intervention," Washington, DC, USA, 2012.*

[5] *J. Piaget, The Origins of Intelligence in Children, New York: International Universities Press, 1952.*

[6] *J. Corrarino, "Disaster-related mental health needs of women and children," MCN AM J Matern Child Nurs, p. 33(4):242–248, 2008.*

[7] *D. Aneelraj, C. N. Kumar, R. Somanathan and e. al., "Uttarakhand Disaster 2013: A Report on Psychosocial Adversities Experienced by Children and Adolescents," Indian J Pediatr, p. 83(4):316–321, 2016.*

[8] *S. A. a. M. H. S. Administration, "Disaster Technical Assistance Center Supplemental Research Bulletin: Behavioral Health Conditions in Children and Youth Exposed to Natural Disasters," 2018.*

[9] *R. Fuchs, M. Glaude, T. Hansel, J. Osofsky and H. Osofsky, "Adolescent risk substance use behavior, posttraumatic stress, depression, and resilience: Innovative considerations for disaster recovery," Subst Abus, p. 42(3):358–365, 2021.*

[10] *B. Tang, X. Liu, Y. Liu, C. Xue and L. Zhang, "A meta-analysis of risk factors for depression in adults and children after natural disasters," BMC Public Health, p. 14(1):1–12, 2014.*

[11] *B. A. v. d. Kolk, The Body Keeps the Score: Brain, Mind, and Body in the Healing of Trauma, New York, NY: Viking Press, 2014.*

[12] *R. L. Aupperle, A. J. Melrose, M. B. Stein and M. P. Paulus, "Executive function and PTSD: disengaging from trauma," Neuropharmacology, p. 62(2):686–694, 2012.*

[13] *M. Merzenich, Soft-Wired: How the New Science of Brain Plasticity Can Change Your Life, San Francisco, CA: Parnassus Publishing, 2013.*

[14] *J. Piaget, The Origins of Intelligence in Children, New York: International Universities Press, 1952.*

[15] *UNESCO, "Teaching and Learning: Achieving Quality for All," EFA Global Monitoring Report, 2014.*

[16] *U. I. f. Statistics, "More Than One-Half of Children and Adolescents Are Not Learning Worldwide," Fact Sheet No. 46, Montreal, Canada, 2017.*

[17] *W. Bank, "Ending Learning Poverty: What Will It Take?," Washington, DC, USA, 2019.*

[18] *C. Blair and R. P. Razza, "Relating effortful control, executive function, and false belief understanding to emerging math and literacy ability in kindergarten," Child Development, p. 78(2):647–663, 2007.*

[19] *I. o. P. S. o. S. Lanka, "Protecting the Education Rights of Children Affected by Displacement," IPS, Colombo, Sri Lanka, 2018.*

[20] *UNESCO, "Education in Emergencies: Global Estimates and Priorities," Paris, France, 2023.*

[21] *H. Crompton, D. Burke, K. Jordan and S. W. G. Wilson, "Learning with technology during emergencies: A systematic review of K–12 education," Br. J. Educ. Technol., p. 52(4):1554–1575, 2021.*

[22] *S. L. R. C. Society, "Situation Report No. 8: Cyclone Ditwah," SLRCS, Colombo, 2025.*

[23] *S. t. Children, "Sri Lanka Cyclone: Tens of Thousands of Children in Temporary Shelters Need Mental Health Support," Save the Children International, London, 2025.*

[24] *OECD, "Jordan: EdTech Without Internet — Kolibri," OECD Education Today, 2021. [Online]. Available: https://oecdedutoday.com/.*

[25] *S. t. C. Australia, "Digital Education in Emergencies: A Lifeline for Learners," Save the Children, 2023.*

[26] *L. Cragg and C. Nation, "Go or no-go? Developmental improvements in the efficiency of response inhibition in mid-childhood," Developmental Science, p. 11(6):819–827, 2008.*

[27] *C. K. Conners, Conners' Continuous Performance Test (CPT), North Tonawanda, NY: Multi-Health Systems, 1994.*

[28] *J. Piaget, "The stages of the intellectual development of the child," Bulletin of the Menninger Clinic, p. 26(3):120–128, 1962.*

[29] *B. S. Bloom, "Learning for Mastery," Evaluation Comment, p. 1(2):1–12, 1968.*

[30] *L. S. Vygotsky, Mind in Society: The Development of Higher Psychological Processes, Cambridge, MA: Harvard University Press, 1978.*

[31] *N. I. o. E. S. Lanka, "National Curriculum Framework for General Education in Sri Lanka," NIE, Maharagama, 2016.*

[32] *F. T. Liu, K. M. Ting and Z.-H. Zhou, "Isolation Forest," in Proc. 8th IEEE Int. Conf. Data Mining (ICDM), Pisa, Italy, 2008.*

[33] *A. H. Maslow, "A Theory of Human Motivation," Psychological Review, p. 50(4):370–396, 1943.*

[34] *UNICEF, "Cyclone Idai: Children Affected in Mozambique, Malawi and Zimbabwe," Theirworld, 2019. [Online]. Available: https://theirworld.org/.*

[35] *P. A. R. Synthesis, "Adaptive Learning in Sri Lanka After a Disaster in a Disaster Relief Camp; ZPD Scaffolding in Digital Education for Disaster Contexts," Research compilation, 2026.*

[36] *J. A. Sergeant, "The cognitive-energetic model: An empirical approach to attention-deficit hyperactivity disorder," Neuroscience & Biobehavioral Reviews, p. 24(1):7–12, 2000.*

[37] *S. K. D'Mello and A. C. Graesser, "Dynamics of affective states during complex learning," Learning and Instruction, p. 22(2):145–157, 2012.*

[38] *V. Chandola, A. Banerjee and V. Kumar, "Anomaly detection: A survey," ACM Computing Surveys, p. 41(3):Art. 15, 2009.*